

How to ^{Edit}~~Write~~ Your Scientific Article

RSV vaccination as an example



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rsmith43.github.io/mysite/paperwritinghandouts.htm

The value of writing

- Writing is the currency of academia

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- With the advent of AI, the tipping point has moved from basic writing to the skill of editing
- Being able to produce well-written, well-argued papers will get you rapidly ahead in your career
 - happily, this is a skill you can learn.

Overview

- What this workshop is not

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1. First Draft

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 6. What's next.

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- Note that this is just one approach
- You may have a different one.

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 - abstract (always leave until last).

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- This is all very standard
- *It does not make your results worth publishing!*

Respiratory Syncytial Virus vaccination

Robert J. Smith¹, Geoff Mercer

Part 2: impulsive model

1 Introduction

2 The model

We extend the basic model from Weber et al to include vaccination.

Assumptions: The leaving rate μ is unchanged across all classes. There is no disease-specific death rate. We scale the entry and leaving rates so that the population is constant.

The basic model with vaccination is

$$\begin{aligned}S' &= \mu(1 - \epsilon p) - \mu S - \beta(t)S(I + I_V) + \gamma R + \omega V \\I' &= \beta(t)S(I + I_V) - \nu I - \mu I + \omega I_V \\R' &= \nu I - \mu R - \gamma R + \omega R_V \\V' &= \epsilon p \mu - \mu V - \beta_V(t)V(I + I_V) + \gamma_V R_V - \omega V \\I'_V &= \beta_V(t)V(I + I_V) - \nu_V I_V - \mu I_V - \omega I_V \\R'_V &= \nu_V I_V - \mu R_V - \gamma_V R_V - \omega R_V\end{aligned}$$

with $\beta(t) = b_0(1 + b_1 \cos(2\pi t + \phi))$ and $\beta_V(t) = (1 - \alpha)\beta(t)$, for $0 \leq \alpha \leq 1$. (We may relax the lower bound on α later.)

3 Analysis

There is a disease-free equilibrium that satisfies

$$(\bar{S}, \bar{I}, \bar{R}, \bar{V}, \bar{I}_V, \bar{R}_V) = \left(\frac{1}{\mu} \left[\mu(1 - \epsilon p) + \frac{\omega \epsilon p \mu}{\mu + \omega} \right], 0, 0, \frac{\epsilon p \mu}{\mu + \omega}, 0, 0 \right)$$

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The core idea.

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Assumptions: homogeneous mixing, no age- or sex-specific classes. There is no disease-specific death rate. We scale the entry and leaving rates so that the population is constant.

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If we assume transmission is constant so that β and β_V are independent of time, then the Jacobian is

$$J = \begin{bmatrix} -\mu - \beta(I + I_V) & -\beta\bar{S} & \gamma & \omega & -\beta\bar{S} & 0 \\ \beta(I + I_V) & \beta\bar{S} - \mu - \nu & 0 & 0 & \beta\bar{S} + \omega & 0 \\ 0 & \nu & -\mu - \gamma & 0 & 0 & \omega \\ 0 & -\beta_V\bar{V} & 0 & -\mu - \beta_V(I + I_V) - \omega & -\beta_V\bar{V} & \gamma_V \\ 0 & \beta_V\bar{V} & 0 & \beta_V(I + I_V) & \beta_V\bar{V} - \nu_V - \mu - \omega & 0 \\ 0 & 0 & 0 & 0 & \nu_V & -\mu - \gamma_V - \omega \end{bmatrix}$$

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The characteristic polynomial satisfies

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The first four eigenvalues are always negative. The nontrivial part of characteristic equation satisfies

$$\lambda^2 + b_1\lambda + c_1 = 0$$

where

$$\begin{aligned} b_1 &= -\beta\bar{S} + \mu + \nu - \beta_V\bar{V} + \nu_V + \mu + \omega \\ c_1 &= (\beta\bar{S} - \mu - \nu)(\beta_V\bar{V} - \nu_V - \mu - \omega) - \beta_V\bar{V}(\beta\bar{S} + \omega) \\ &= \beta\bar{S}(-\nu_V - \mu - \omega) - (\mu + \nu)(\beta_V\bar{V} - \nu_V - \mu - \omega) - \beta_V\bar{V}\omega \end{aligned}$$

From $c_1 = 0$, we find

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(This is equivalent to the value found using the next-generation method.)

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Standard but
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- Key question: What do we learn that we didn't know?

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- Key question: What do we learn that we didn't know?
 - it's possible to destabilise the equilibrium in a new way (not a backward bifurcation)
- There's no algorithm for this part.

If $c_1 = 0$ and $b_1 > 0$, then we have a bifurcation point with the property that the DFE is stable if $R_0 < 1$ and unstable if $R_0 > 1$

However, it is possible that when $c_1 = 0$, $b_1 < 0$. In this case, R_0 is not a threshold and the disease can persist if $R_0 < 1$.

When $c_1 = 0$, we have

$$b_1 \Big|_{c_1=0} = \frac{1}{\nu_V + \mu + \omega} [\beta_V \bar{V}(\nu - \nu_V) + (\nu_V + \mu + \omega)^2]$$

Note that if $\nu = \nu_V$, then $b_1 > 0$. However, we expect that vaccinated individuals will recover faster than unvaccinated individuals. Thus $\nu_V > \nu$. This raises the possibility that b_1 could be negative.

if $\nu_V \rightarrow \infty$, then this is equivalent to vaccinated individuals recovering instantaneously. In this case,

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Surprising result

If $c_1 = 0$ and $b_1 > 0$, then we have a bifurcation point with the property that the DFE is stable if $R_0 < 1$ and unstable if $R_0 > 1$

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May not happen...

Note that if $\nu = \nu_V$, then $b_1 > 0$. However, we expect that vaccinated individuals will recover faster than unvaccinated individuals. Thus $\nu_V > \nu$. This raises the possibility that b_1 could be negative.

if $\nu_V \rightarrow \infty$, then this is equivalent to vaccinated individuals recovering instantaneously. In this case,

$$\begin{aligned} \lim_{\nu_V \rightarrow \infty} b_1 &= \lim_{\nu_V \rightarrow \infty} \frac{\beta_V \bar{V}(\nu - \nu_V)}{\omega + \mu + \nu_V} + \omega + \mu + \nu_V \\ &= \infty - \beta_V \bar{V} > 0 \end{aligned}$$

Surprising result

If $c_1 = 0$ and $b_1 > 0$, then we have a bifurcation point with the property that the DFE is stable if $R_0 < 1$ and unstable if $R_0 > 1$

However, it is possible that when $c_1 = 0$, $b_1 < 0$. In this case, R_0 is not a threshold and the disease can persist if $R_0 < 1$.

When $c_1 = 0$, we have

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Extreme case.

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Hence if we define $f(\nu_V) = \frac{\beta_V \bar{V}(\nu - \nu_V) + (\omega + \mu + \nu_V)^2}{\omega + \mu + \nu_V}$, then we would like to know whether f has a turning point ν_V^* such that $f(\nu_V^*) < 0$.

We have

$$\begin{aligned} f'(\nu_V) &= \frac{(\omega + \mu + \nu_V)[- \beta_V \bar{V} + 2(\omega + \mu + \nu_V)] - [\beta_V \bar{V}(\nu - \nu_V) + (\omega + \mu + \nu_V)^2]}{(\omega + \mu + \nu_V)^2} \\ &= \frac{(\omega + \mu + \nu_V)^2 - \beta_V \bar{V}[\omega + \mu + \nu]}{(\omega + \mu + \nu_V)^2} \end{aligned}$$

It follows that $\nu_V^* = \sqrt{\beta_V \bar{V}(\omega + \mu + \nu)} - \omega - \mu$.

There are two requirements we need for this to be meaningful: 1. $\nu_V^* > \nu$ and 2. $f(\nu_V^*) < 0$. See Figure 1.

[Is this definitely a local minimum?]

Yes, proven. Write up results.

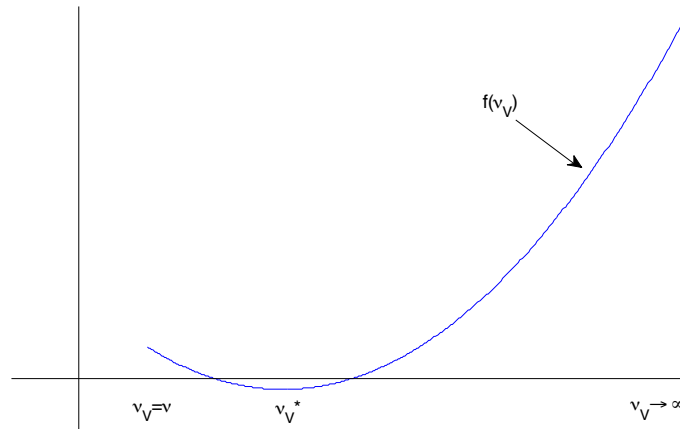


Figure 1: Possible sketch of the form of $f(\nu_V)$ with a negative minimum between two positive extremes.

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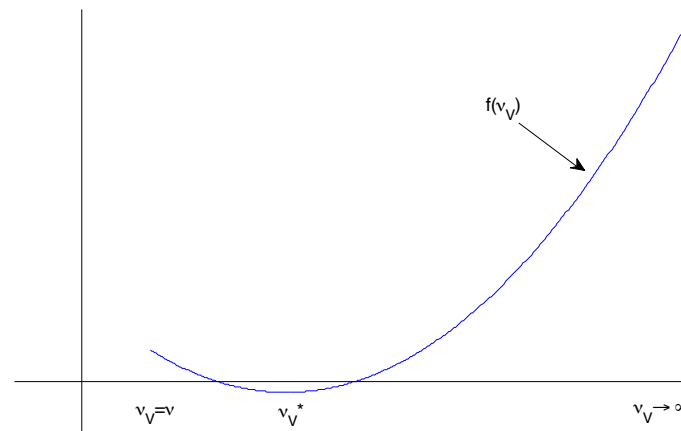


Figure 1: Possible sketch of the form of $f(\nu_V)$ with a negative minimum between two positive extremes.

Identify and
solve key issues.

Further ideas

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- *Superficial or single results should not be published.*

4 Impulsive model

$$S' = \mu - \mu S - \beta(t)S(I + I_V) + \gamma R + \omega V \quad t \neq t_k$$

$$I' = \beta(t)S(I + I_V) - \nu I - \mu I + \omega I_V \quad t \neq t_k$$

$$R' = \nu I - \mu R - \gamma R + \omega R_V \quad t \neq t_k$$

$$V' = -\mu V - \beta_V(t)V(I + I_V) + \gamma_V R_V - \omega V \quad t \neq t_k$$

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- Note that this is the entirety of Section 4 in the first draft.

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 - extreme parameters can induce infection spikes, even with 100% vaccination.

5 Numerical simulations

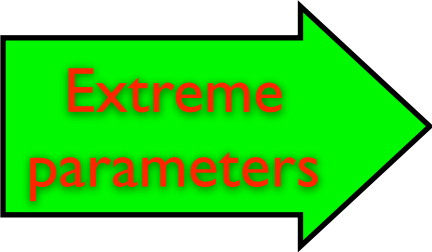
From Weber et al, we have $\beta = 0.03$, $\mu = 0.041$ and $\nu = 36$. We add vaccination parameters $\omega = 0.1$, $\epsilon = 1$, $p = 1$, $\nu_V = 177$ and $\beta_V = 3000$. (We also have $\gamma = 1.8$ and impose $\gamma_V = 1.2\gamma$.) This represents a vaccine with complete coverage and perfect efficacy that wanes after ten years, but vaccinated individuals can be infected with a high transmission rate, but recover very quickly.

Although the transmission rate is unrealistically high, this nevertheless demonstrates that a stable DFE can be destabilised by a vaccine.

Note that what we are dealing with here is not a backward bifurcation, but rather a destabilisation of the equilibrium.

Figure 2 shows the results of transmission using data from Weber et al and assumed vaccination parameters such that recovery was slightly faster and transmission slightly less likely. The vaccine was given to 50% of the eligible population, but waned after 0.01 years (**check this**) The data used were $\mu = 0.041$; $\omega = 100$; $\beta = 50$; $\beta_V = 0.8\beta$; $\epsilon = 0.9$; $p = 0.5$; $\nu = 36$; $\nu_V = 1.2\nu$; $\gamma = 1.8$; $\gamma_V = 1.2\gamma$.

Figure 3 illustrates the destabilisation of the DFE when extreme vaccination parameters are used. In this case, transmission of the vaccinated strain was extremely high but recovery extremely fast, allowing for infection spikes to occur among a small proportion of vaccinated individuals before the infection stabilises. Data used were $\mu = 0.041$; $\omega = 0.1$; $\beta = 0.03$; $\beta_V = 3000$; $\epsilon = 1$; $p = 0, 1$; $\nu = 36$; $\nu_V = 177$; $\gamma = 1.8$; $\gamma_V = 1.2\gamma$.



Extreme
parameters

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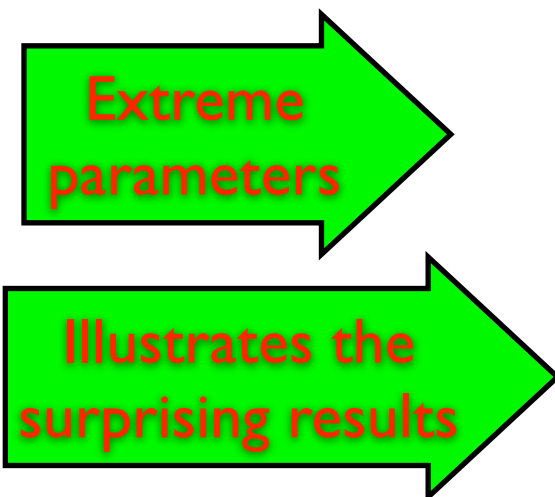
From Weber et al, we have $\beta = 0.03$ [26]. We add vaccination parameters $\omega = 0.1$, $\epsilon = 1$, $p = 0.5$, $\nu_V = 177$ and $\beta_V = 3000$. We also have $\gamma = 1.8$ and impose $\gamma_V = 1.2\gamma$.) This represents a vaccine with complete coverage and perfect efficacy that wanes after ten years, but vaccinated individuals can be infected with a high transmission rate, but recover very quickly.

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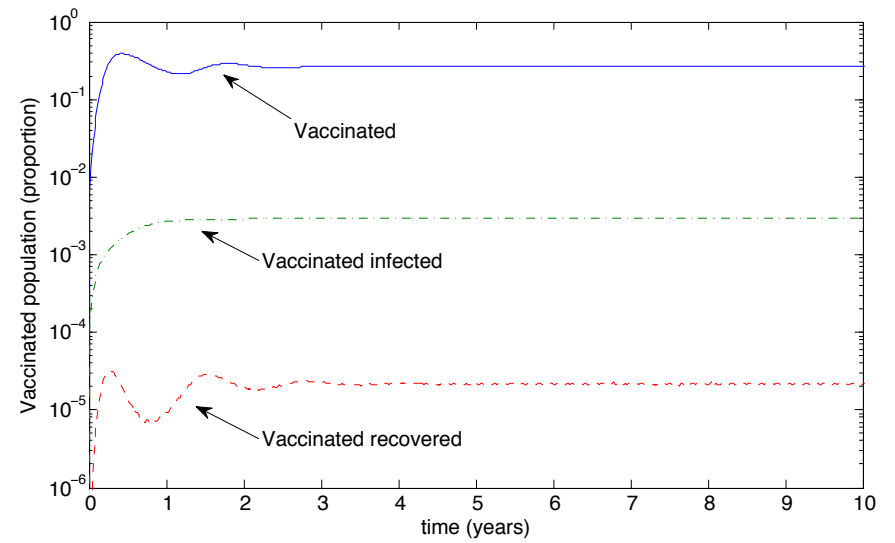
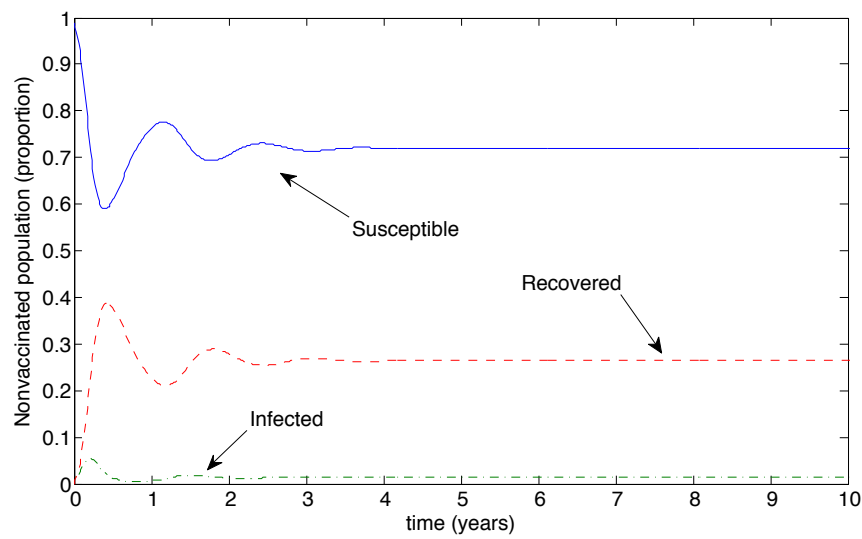


Figure 2: Results from the basic model with vaccination. There is an outbreak and the disease oscillates, eventually approaching an equilibrium. A small proportion of individuals are (and remain) vaccinated, with a low-level outbreak among vaccinated individuals.

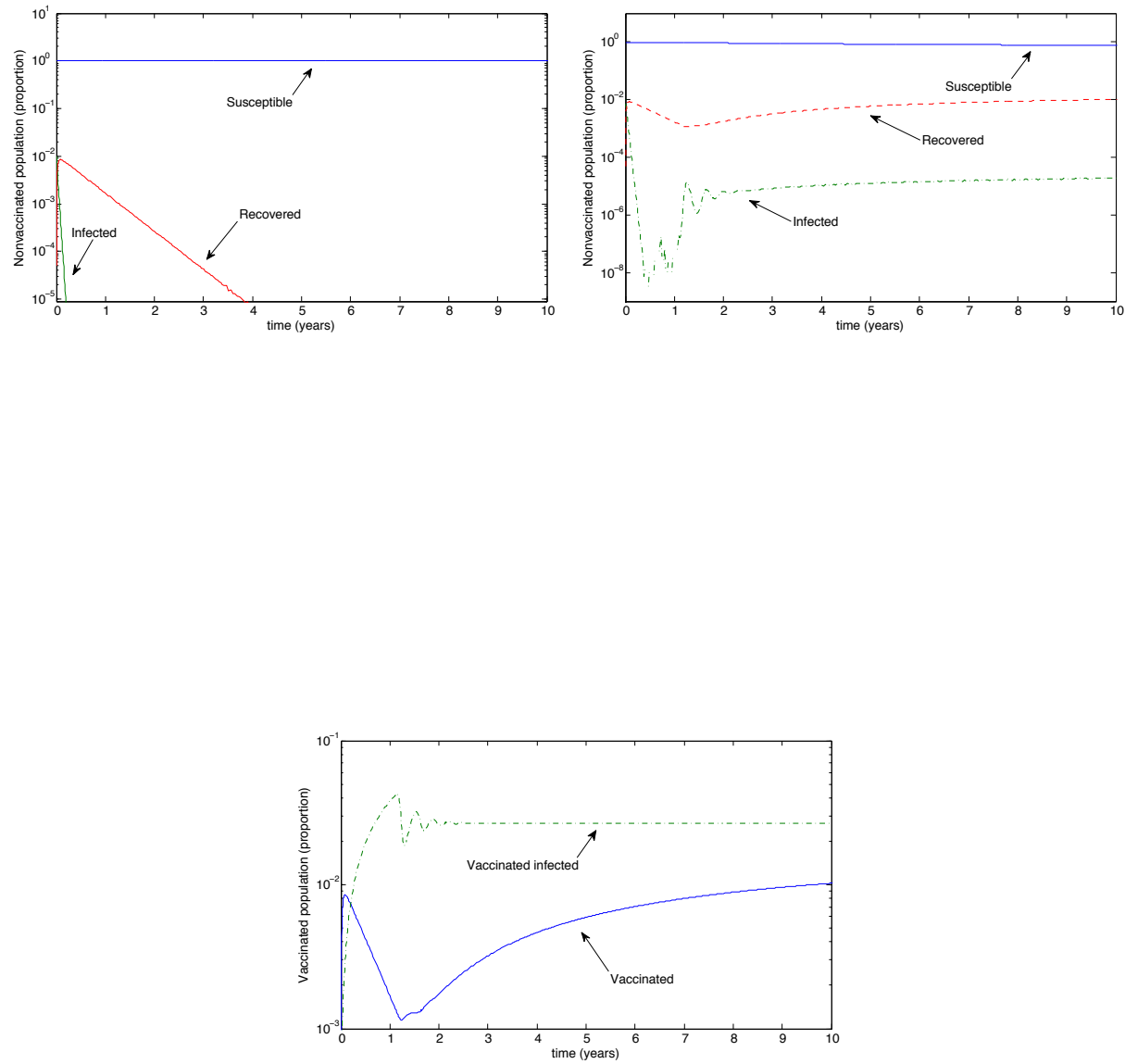
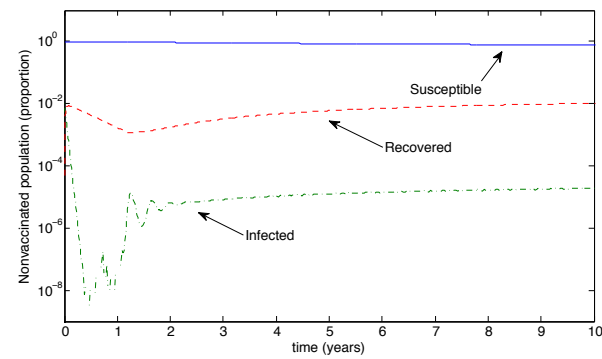
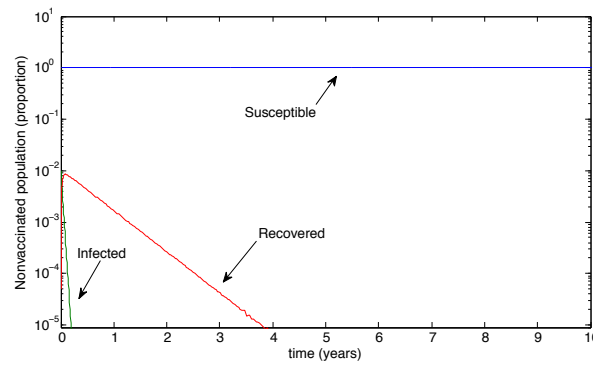


Figure 3: Extreme parameters show that perfect vaccination can induce infection spikes. A. With no vaccine, the result is that the infection clears and the entire population remains susceptible (note that the low-level fluctuations result from numerical limitations in MATLAB) B. With a vaccine given to the entire population, the susceptible population dips slightly as infection takes hold. C. Infection in the vaccinated population initially takes the form of infection spikes before stabilising. Note that vaccination thus destabilises the disease-free equilibrium.



Infection spikes.

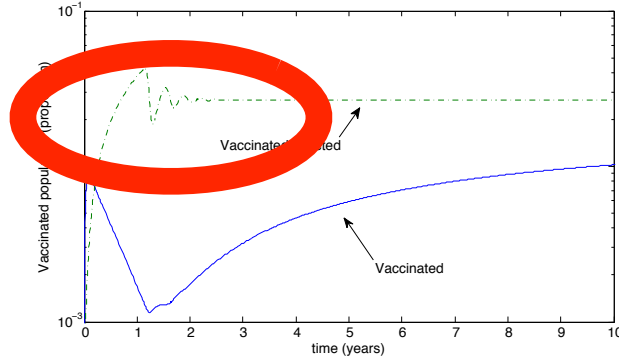


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- Identify key features
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 - these should be issues people want to know.

Next, following Weber et al, we examined the more realistic case when the transmission rate oscillated. Since the waning rate of the vaccine was not known, we decided to investigate several options for ω .

When there is no vaccine, the disease results in a maximum of 7% of the population infected. Data used was $\mu = 1/70$; $\omega = 1/10$; $b_0 = 60$; $b_1 = 0.16$; $\phi = 0.15$; $\beta_V = 0.5\beta$; $\epsilon = 1$; $p = 0$; $\nu = 36$; $\nu_V = 1.2\nu$; $\gamma = 1.8$; $\gamma_V = 1.2\gamma$. See Figure 4.

A vaccine given to the entire population with 50% transmission that did not wane for ten years resulted in about 6% of the population infected. Data used was identical to Figure 4 except that $p = 1$. See Figure 5. In this case, there is only a slight decrease in the maximum disease burden, despite complete vaccination coverage.

[If the entire population is vaccinated, who is susceptible?]

A vaccine given to the entire population with 50% transmission that did not wane for 70 years resulted in a significant reduction in the infected population. Data used was identical to Figure 5 except that $\omega = 1/70$. See Figure 6. In this case, there is a significant reduction in the total disease burden, reducing the maximum to less than 2% of the total population.

Of course, complete vaccination coverage is not realistic. Consequently, we examined the effect of 50% coverage with a vaccine that did not wane for 70 years. Data used was identical to Figure 6 except that $p = 0.5$. See Figure 7. In this case, there is still a significant reduction in total disease burden. Note that significantly greater reduction is achieved with 50% coverage and a lifelong vaccine than was achieved with 100% coverage and a vaccine that lasted 10 years (see Figure 5).

It follows that the waning rate of the vaccine is crucial. Even if complete coverage could be achieved, a vaccine with a moderate duration (eg 10 years) results in very little reduction of infection. Conversely, a vaccine that does not wane over a lifetime results in significant reduction in disease burden.

The best-case scenario involves complete coverage with a vaccine that does not wane for 70 years. Figure 8 illustrates the population dynamics when such a vaccine is introduced.



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A vaccine given to the entire population with 50% transmission that did not wane for ten years resulted in about 6% of the population infected. Data used was identical to Figure 4 except that $p = 1$. See Figure 5. In this case, there is only a slight decrease in the maximum disease burden, despite complete vaccination coverage.

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Of course, complete vaccination coverage is not realistic. Consequently, we examined the effect of 50% coverage with a vaccine that did not wane for 70 years. Data used was identical to Figure 6 except that $p = 0.5$. See Figure 7. In this case, there is still a significant reduction in total disease burden. Note that significantly greater reduction is achieved with 50% coverage and a lifelong vaccine than was achieved with 100% coverage and a vaccine that lasted 10 years (see Figure 5).

It follows that the waning rate of the vaccine is crucial. Even if complete coverage could be achieved, a vaccine with a moderate duration (eg 10 years) results in very little reduction of infection. Conversely, a vaccine that does not wane over a lifetime results in significant reduction in disease burden.

The best-case scenario involves complete coverage with a vaccine that does not wane for 70 years. Figure 8 illustrates the population dynamics when such a vaccine is introduced.

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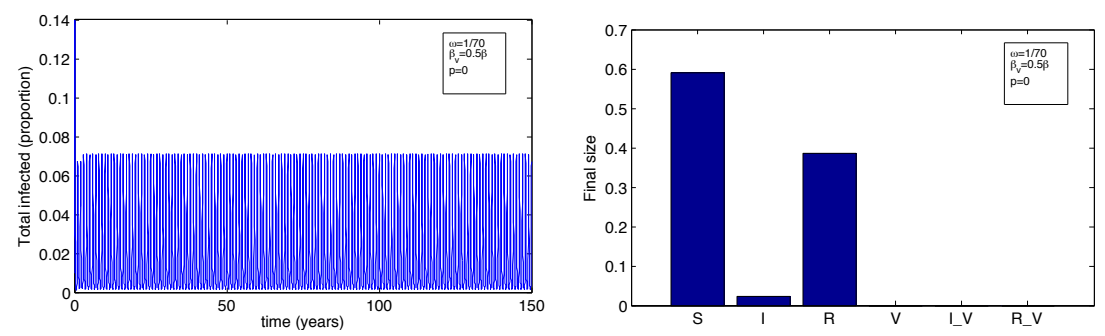


Figure 4: Without vaccination, the disease infects up to 7% of the population. A. The total infected population, including vaccinated individuals. B. The final size in each population.

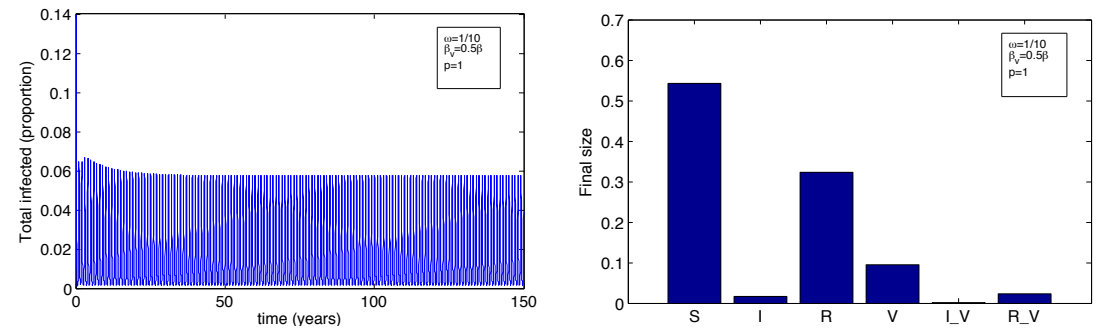


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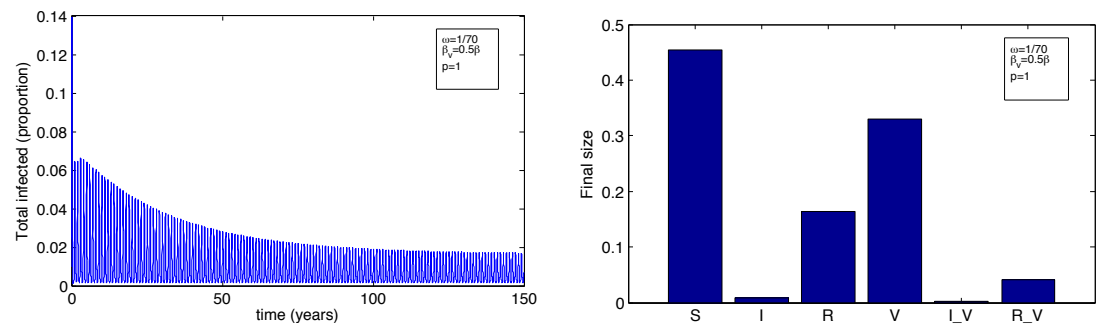


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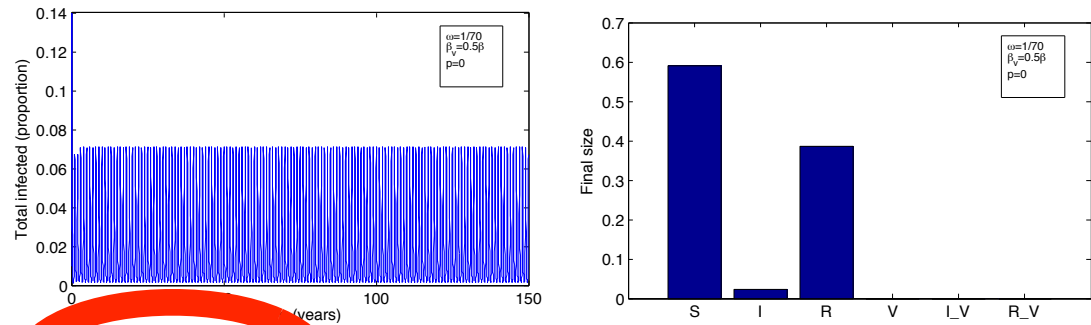


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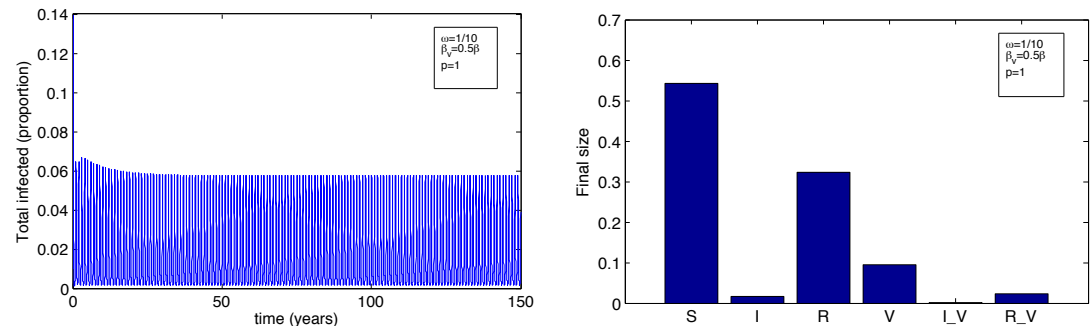


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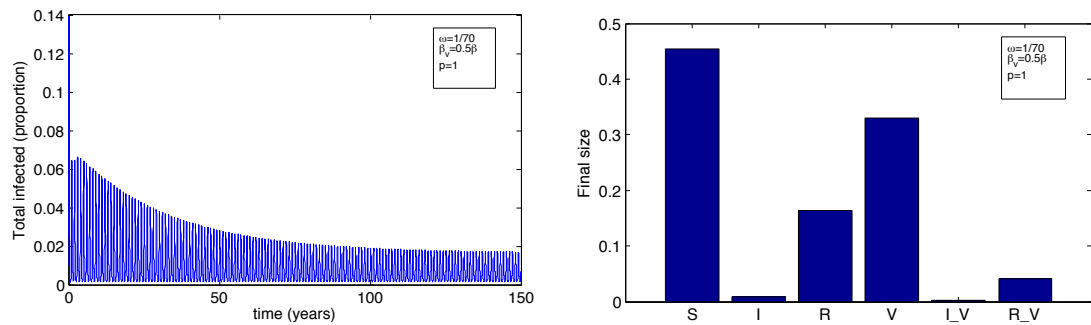


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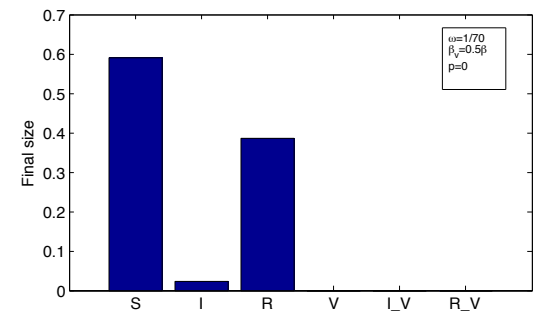
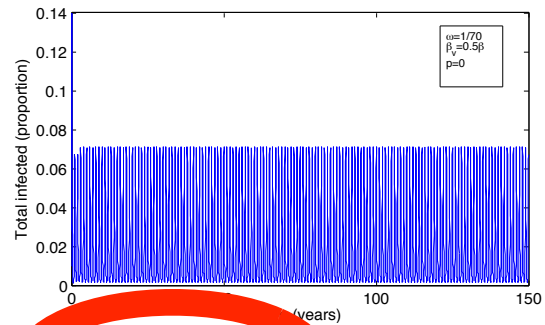


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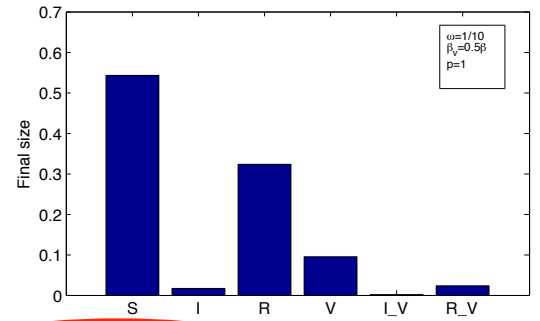
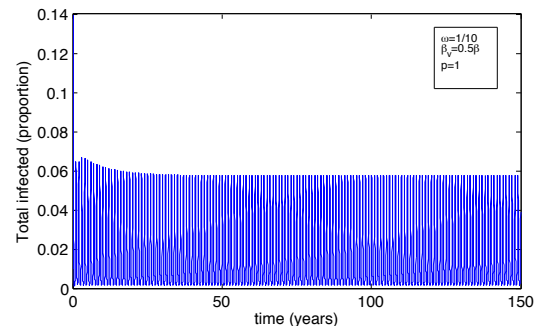


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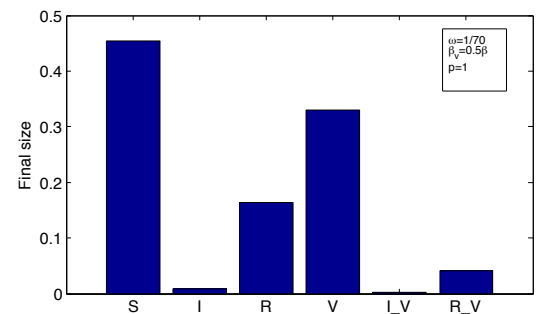
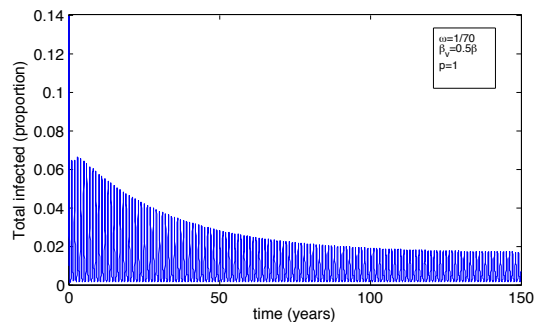


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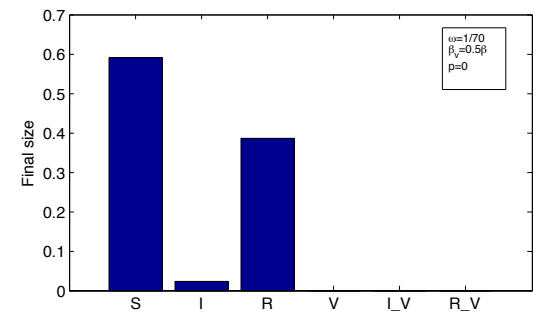
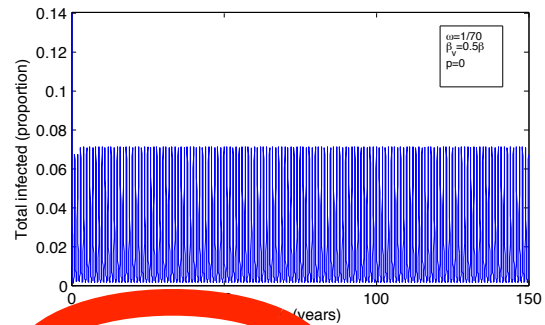


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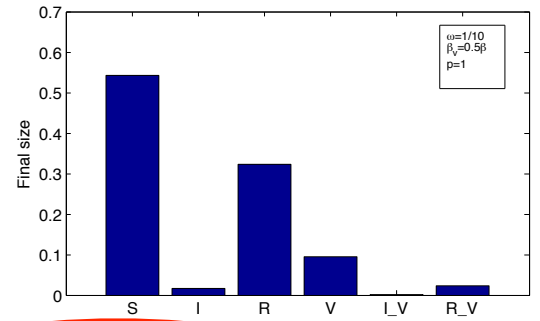
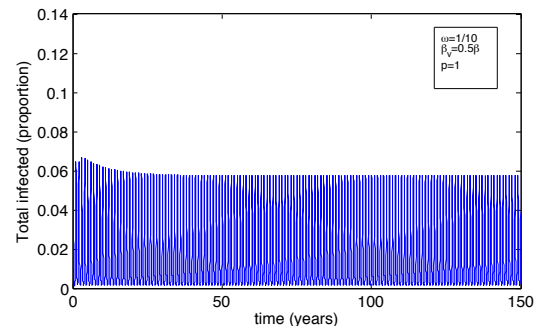


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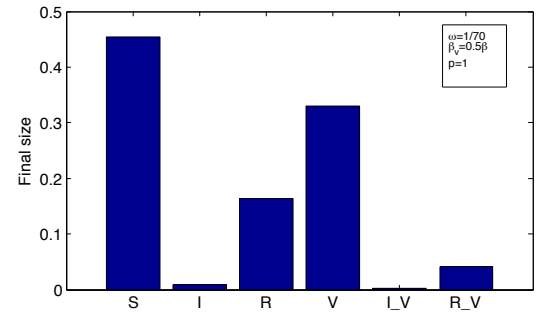
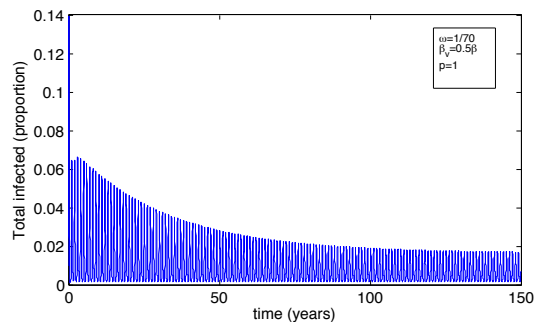


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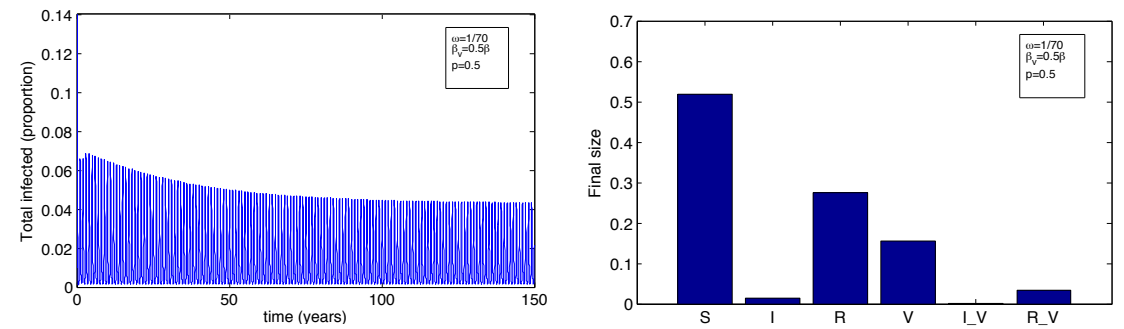


Figure 7: 50% coverage with a vaccine that did not wane for 70 years results in a significant reduction in infection. A. The total infected population, including vaccinated individuals. B. The final size in each population.

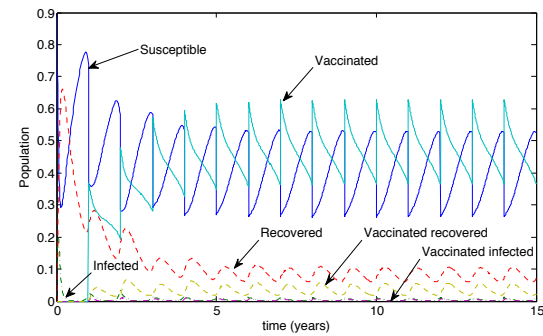


Figure 8: Population dynamics for a lifelong vaccine with complete coverage. Note that the vaccinated infected are too small to appear on the figure. **[Fix figure; the lowest arrow is regular infected.]**

Different case

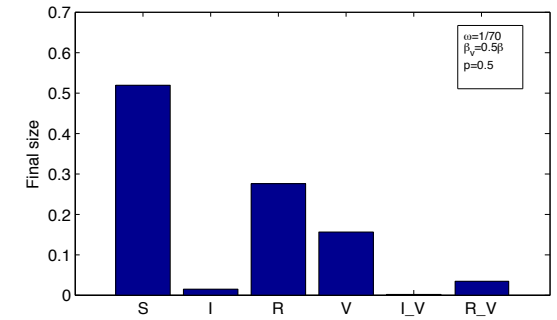
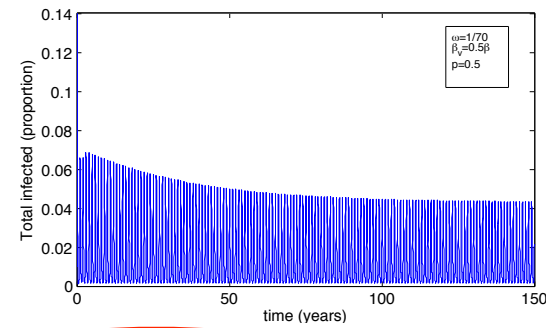


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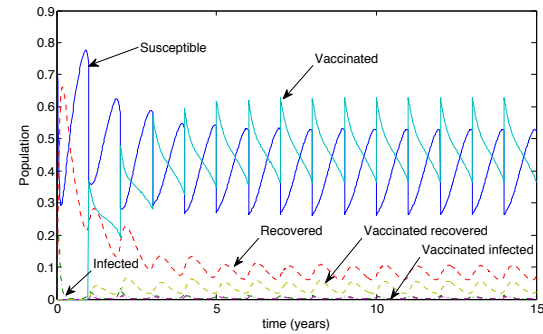


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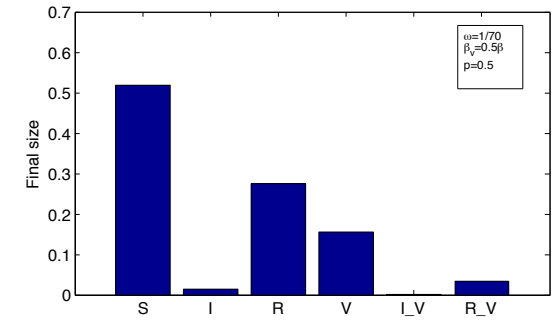
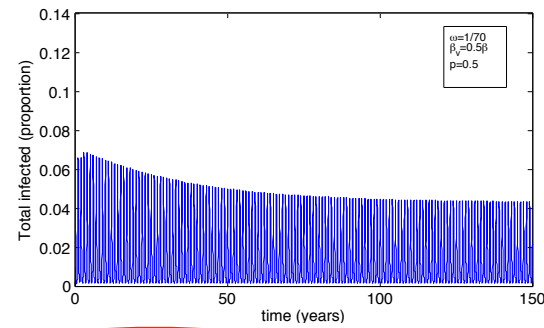


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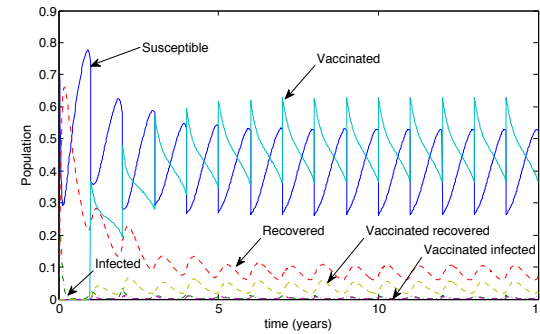


Figure 8: Population dynamics for 100% coverage. Note that the vaccinated infected are too small to appear on the figure. [Fix figure; the lower arrow is regular infected.]

Identify issues.

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I. INTRODUCTION

Respiratory syncytial virus (RSV) is the main cause of acute lower respiratory infections in infants and young children [22], with almost all children having been infected by two years of age [10, 25] and an estimated 0.5–2% of infants requiring hospitalisation due to infection [18]. One recent study estimated that in 2005, 33.8 million new episodes of RSV occurred worldwide in children younger than five years of age [22]. Symptoms of RSV range from those of a cold, more severe afflictions such as bronchiolitis and pneumonia [10]. While mortality due to RSV infection in developed countries is low, occurring in less than 0.1% of cases [32], little data is published about RSV morbidity and mortality in developing countries [34]. However, estimates of the hospitalisation costs are substantial [14, 30, 36], making RSV a significant economic and health care system burden.

Newborn infants are typically protected from RSV infection by maternal antibodies until about six weeks of age [8], and the highest number of observed RSV cases occur in children aged six weeks to six months [5, 27]. Immunity to RSV following an infection is short-lasting and reinfection in childhood is common [19]. Few studies have been undertaken to investigate transmission of RSV among adults, but it is thought that infection can occur throughout life [6, 15] and that in older children and adults, RSV manifests as a mild cold [10, 16]. RSV has been identified as a cause of mortality in the elderly with documented outbreaks in aged care settings [13, 31]; one such study found that up to 18% of pneumonia hospitalisation in adults aged above 65 years may be due to RSV infection [12].

In temperate climates RSV epidemics exhibit distinct and consistent seasonal patterns. Most RSV infections occur during the cooler winter months, whether wet or dry [34], and outbreaks typically last between two and five months [11, 23]. In a number of temperate regions a biennial pattern for RSV cases has been identified; see, for example, [4, 20, 28]. In tropical climates RSV is detected throughout the year with less pronounced seasonal peaks, and the onset of RSV is typically associated with the wet season [26, 34].

Immunoprophylaxis with the monoclonal antibody Palivizumab, while not preventing the onset of infection, has proven effective in reducing the severity of RSV-related symptoms [29]. However, prophylaxis is expensive and generally only administered to high-risk children, with recommendations varying across jurisdictions. There is currently no licensed vaccine to prevent RSV infection, despite about 50 years of vaccine research. Recent research has focused on the development of live attenuated vaccines; several such vaccines are being evaluated in clinical trials, with other vaccines in preclinical development [9, 14]. With the possibility of a RSV vaccine becoming available, mathematical models can be powerful tools for planning vaccination roll-out strategies.

Several ordinary differential equation Susceptible-Exposed-Infectious-Recovered (SEIR) type mathematical models for RSV transmission have been published to date, such as those presented in [3, 7, 17, 21, 24, 33, 35] with a sine or cosine forcing term to account for seasonal variation in transmission. Weber *et al.* [33] present a SEIRS model which incorporates a gradual reduction in susceptibility to reinfection and maternally derived immunity, and fit the model to several data sets. Leecaster *et al.* [17] present a SEIDR model with both child and adult classes for the S, E and I compartments, and where the D class represents children in which

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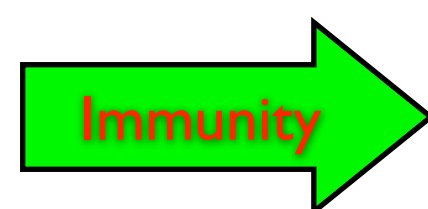
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Respiratory syncytial virus (RSV) is the main cause of acute lower respiratory infections in infants and young children [22], with almost all children having been infected by two years of age [10, 25] and an estimated 0.5–2% of infants requiring hospitalisation due to infection [18]. One recent study estimated that in 2005, 33.8 million new episodes of RSV occurred worldwide in children younger than five years of age [22]. Symptoms of RSV range from those of a cold, more severe afflictions such as bronchiolitis and pneumonia [10]. While mortality due to RSV infection in developed countries is low, occurring in less than 0.1% of cases [32], little data is published about RSV morbidity and mortality in developing countries [34]. However, estimates of the hospitalisation costs are substantial [14, 30, 36], making RSV a significant economic and health care system burden.



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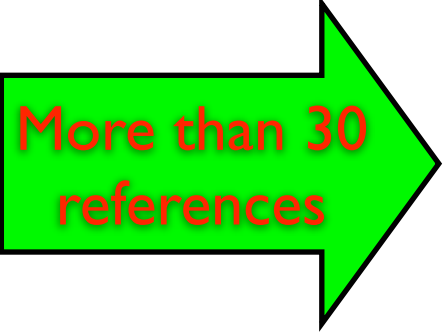
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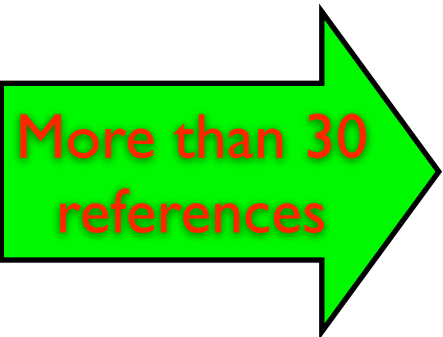
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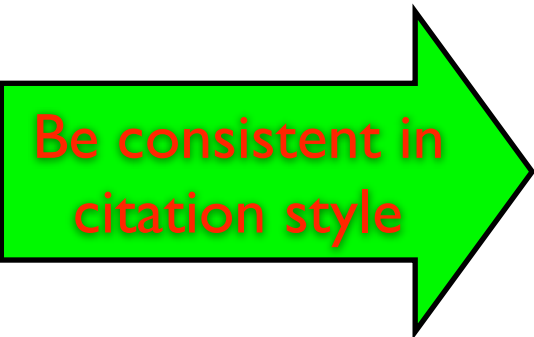
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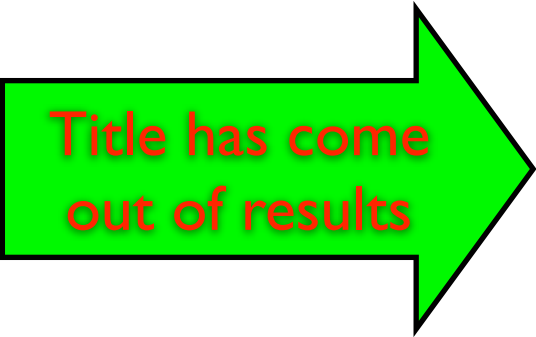
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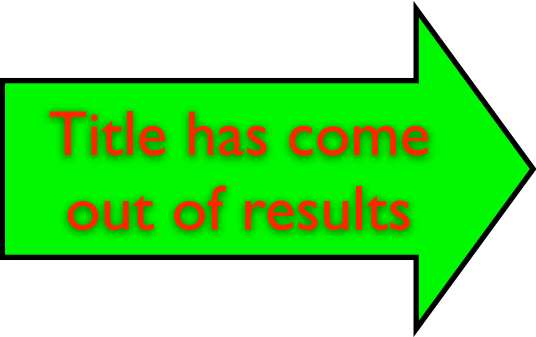
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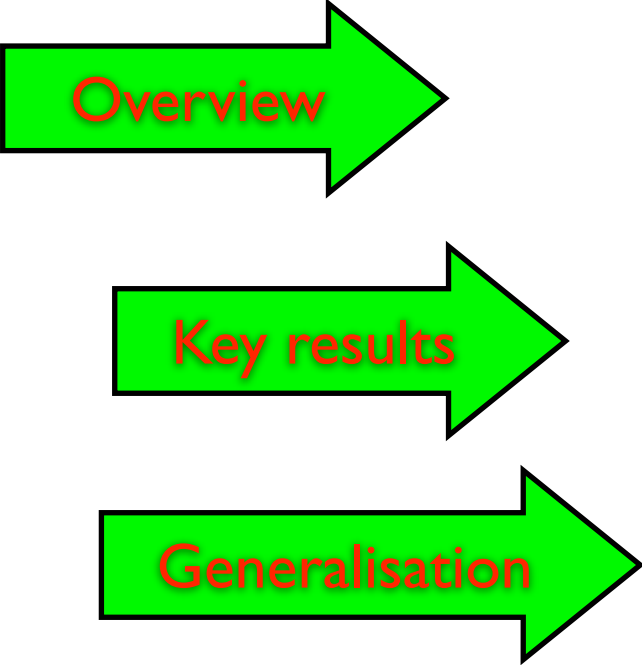
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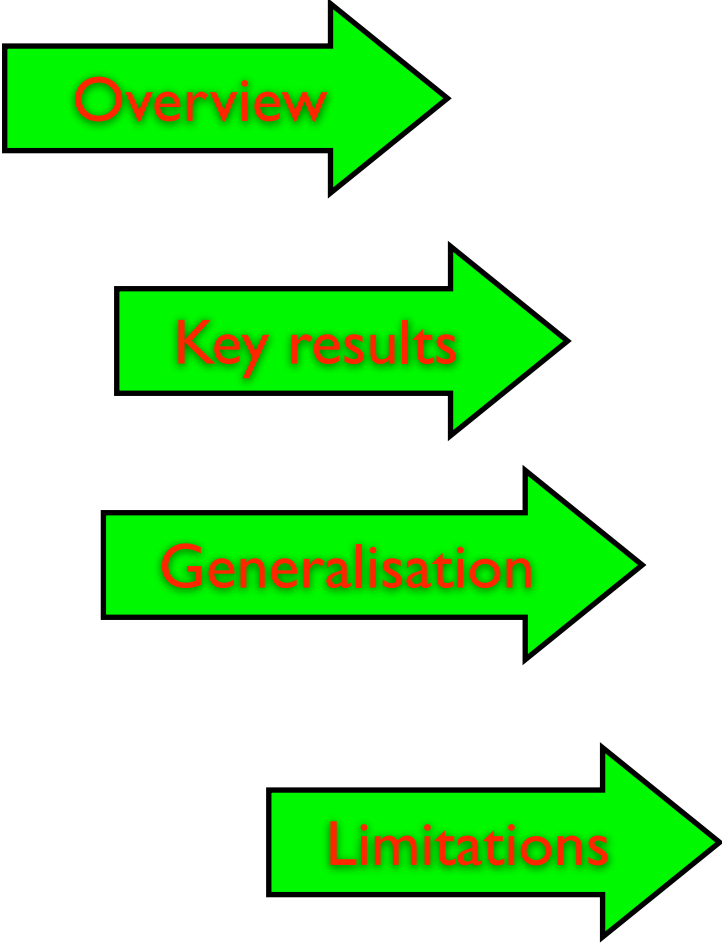
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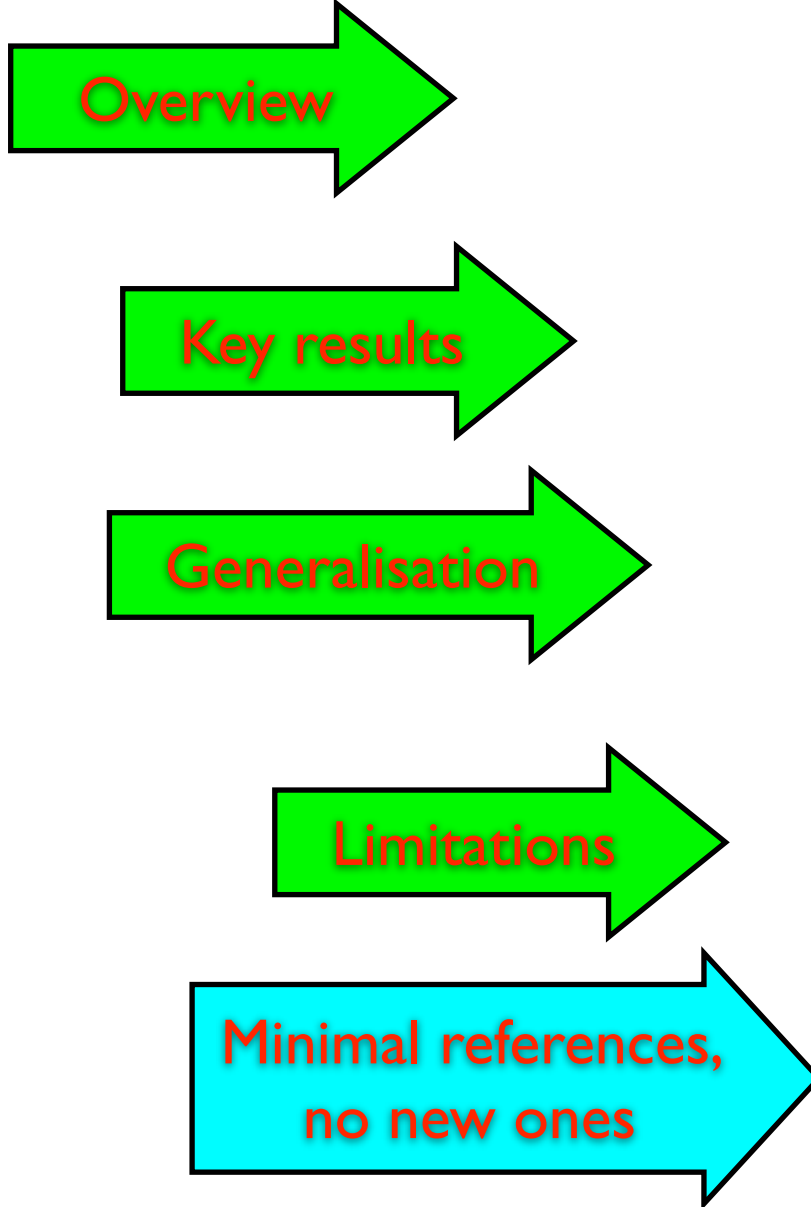
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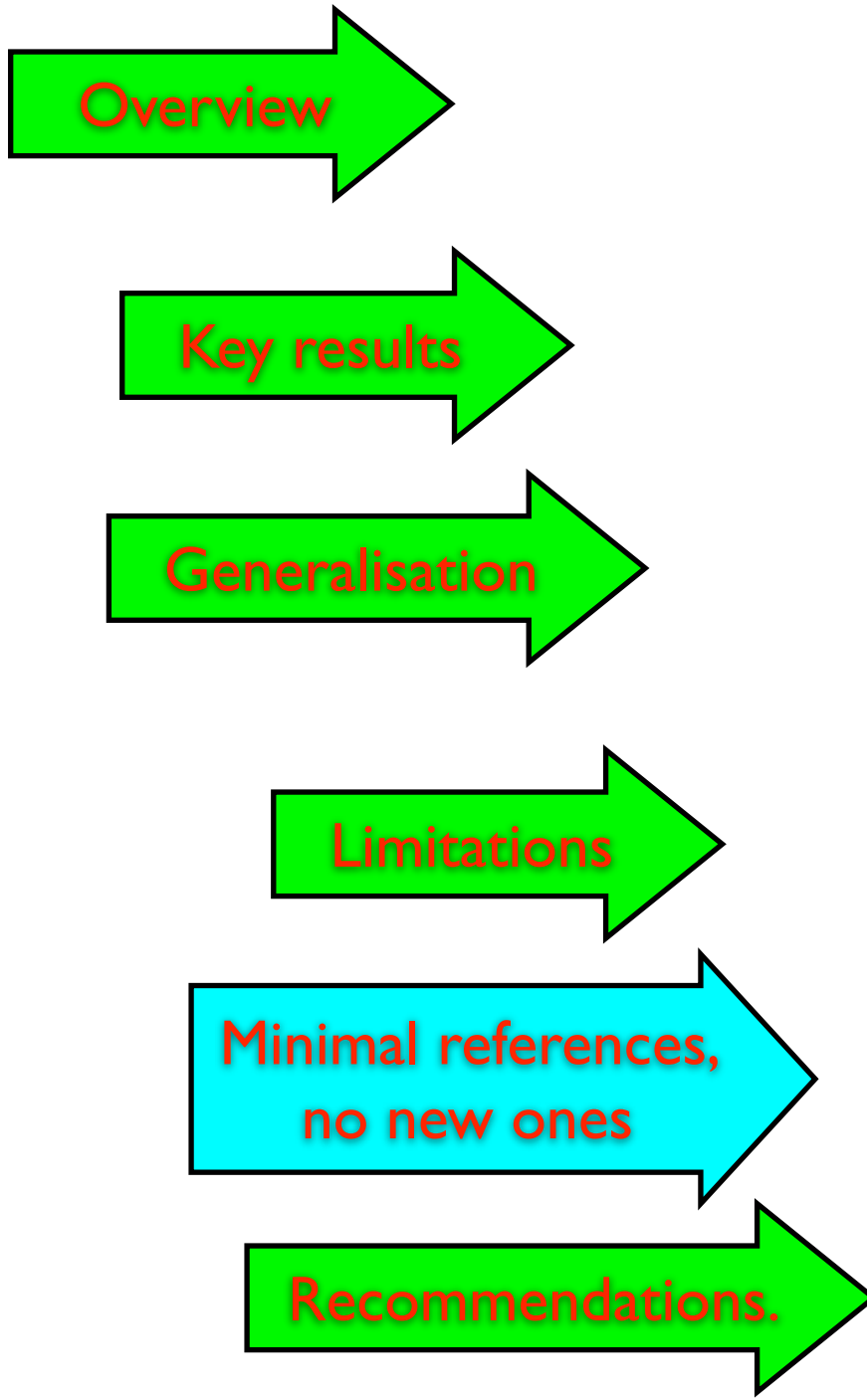
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Our model has some limitations, which should be acknowledged. We assumed that time to administer the vaccine was significantly shorter than the time between vaccine administrations in order to justify the impulsive approximation. Such assumptions are reasonable in many cases [30], although can produce confounding effects in some situations [11]. The extreme parameters that we used to illustrate the vaccination spikes operated under the assumption that the transmission rate for infected vaccinated individuals was significantly higher than the transmission rate without vaccination. Since we extended the model of Weber *et al.* [39], our model inherited many of the assumptions from that model, such as mass-action transmission, a constant birth rate and that the birth and death rates were matched, resulting in a constant population.

A vaccine that targets RSV infection has the potential to significantly reduce the overall prevalence of the disease, but it has to be sufficiently long-lasting. Coverage and effectiveness of the vaccine is important, but the critical parameter that our modelling identified is the waning rate of the vaccine. We thus recommend that candidate vaccines be tested for sufficient duration before being released to the public. If a durable vaccine can be developed, then we stand a chance of controlling this disease, assuming sufficiently widespread coverage.



Part 4: Refinement

- This is where your article really begins!

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- What problems would a reviewer notice?

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While mortality due to RSV infection in developed countries is low, occurring in less than 0.1% of cases [38], few data have been published about RSV morbidity and mortality in developing countries [40]. However, estimates of the hospitalisation costs are substantial [14, 36, 42], making RSV a significant economic and health care system burden.

Newborn infants are typically protected from RSV infection by maternal antibodies until about six weeks of age [9], and the highest number of observed RSV cases occur in children aged six weeks to six months [6, 33]. Immunity to RSV following an infection is short-lasting, and reinfection in childhood is common [22]. Few studies have been undertaken to investigate transmission of RSV among adults, but it is thought that infection can occur throughout life [7, 15] and that, in older children and adults, RSV manifests as a mild cold [10, 18]. RSV has been identified as a cause of mortality in the elderly, with documented outbreaks in aged-care settings [13, 37]; one such study found that up to 18% of pneumonia hospitalisation in adults aged above 65 years may be due to RSV infection [12] **could take out this paragraph?*.

In temperate climates RSV epidemics exhibit distinct and consistent seasonal patterns. Most RSV infections occur during the cooler winter months, whether wet or dry [40], and outbreaks typically last between two and five months [11, 26]. In a number of temperate regions, a biennial pattern for RSV cases has been identified [4, 23, 34]. In tropical climates, RSV is detected throughout the year with less pronounced seasonal peaks, and the onset of RSV is typically associated with the wet season [32, 40].

Immunoprophylaxis with the monoclonal antibody Palivizumab, while not preventing the onset of infection, has proven effective in reducing the severity of RSV-related symptoms [35]. However, prophylaxis is expensive and generally only administered to high-risk children, with recommendations varying across jurisdictions. There is currently no licensed vaccine to prevent RSV infection, despite about 50 years of vaccine research. *Recent research has focused on the development of particle-based and subunit vaccines; several such vaccines are being evaluated in clinical trials, with other vaccines in preclinical development [28, 30]. With the possibility of an RSV vaccine becoming available, mathematical models are powerful tools for assessing the impacts of different vaccine characteristics.*

Several ordinary differential equation mathematical models for RSV transmission have been published to date, most using Susceptible–Exposed–Infectious–Recovered (SEIR) dynamics and with a sine or cosine forcing term to account for seasonal variation in transmission [3, 8, 19, 24, 27, 39, 41]. *Few*



Don't be afraid to
make big cuts

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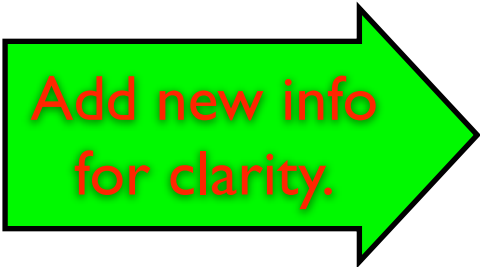
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Add new info
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Here, we examine the effects of a theoretical vaccine on the transmission of RSV in a single age class. We consider several vaccination scenarios, including differing levels of coverage, seasonal oscillations in the transmission rate and a waning of the vaccine. We also compare continuous vaccination to impulsive vaccination in order to determine conditions on the vaccination strength and duration? that will control the virus.

2 The model

We extend the basic compartmental model for a single age cohort described by Weber *et al.* [39] to include a vaccine strategy for RSV where a fixed proportion of newborns are vaccinated?. We assume that the leaving rate μ is unchanged across all classes and that there is no disease-specific death rate. We scale the entry and leaving rates so that the population is constant.

Let S represent susceptible, I represent infected and R represent recovered individuals, with V , I_V and R_V the corresponding compartments for vaccinated individuals. The birth rate is μ , with a proportion p vaccinated, of whom ϵ successfully mount an immune response; the death rate is equal to the birth rate. The time-dependent transmissibility function is $\beta(t)$, with recovery ν and loss of immunity γ . The transmissibility of infected vaccinated individuals is described by $\beta_V(t)$, and the recovery and loss of immunity rates for vaccinated individuals are ν_V and γ_V respectively. Finally, the waning of the vaccine protectiveness? is given by ω .



Has more research
come to light?

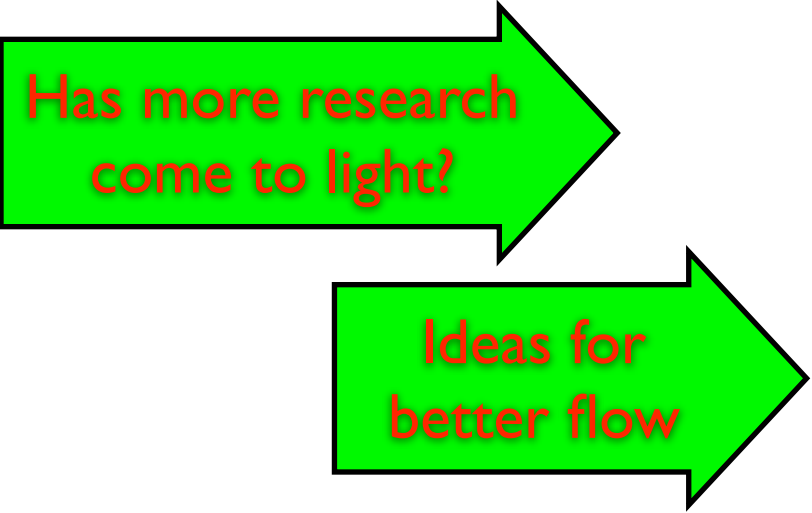
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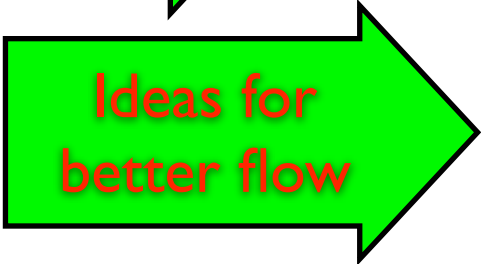
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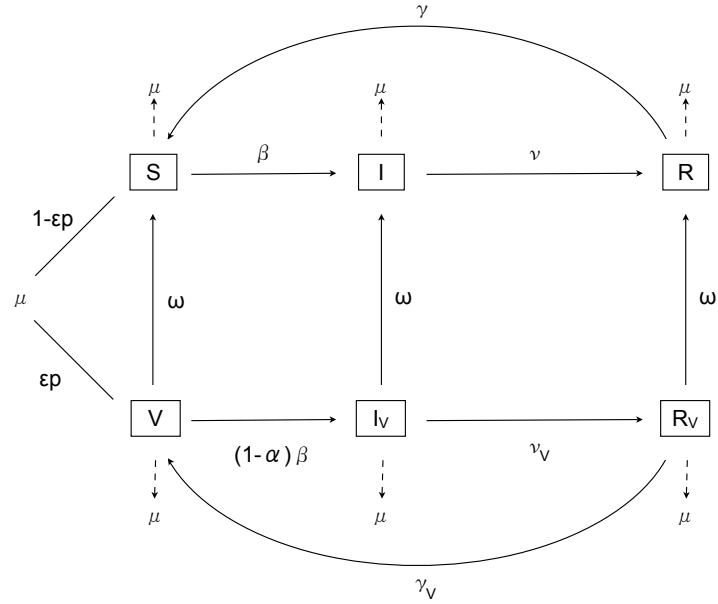


Figure 1: The model.

The basic model with vaccination is then

$$\begin{aligned}
 S' &= \mu(1 - \epsilon p) - \mu S - \beta(t)S(I + I_V) + \gamma R + \omega V \\
 I' &= \beta(t)S(I + I_V) - \nu I - \mu I + \omega I_V \\
 R' &= \nu I - \mu R - \gamma R + \omega R_V \\
 V' &= \epsilon p \mu - \mu V - \beta_V(t)V(I + I_V) + \gamma_V R_V - \omega V \\
 I_V' &= \beta_V(t)V(I + I_V) - \nu_V I_V - \mu I_V - \omega I_V \\
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 \end{aligned}$$

with $\beta(t) = b_0(1 + b_1 \cos(2\pi t + \phi))$ and $\beta_V(t) = (1 - \alpha)\beta(t)$, for $0 \leq \alpha \leq 1$, where α represents.... (We may relax the lower bound on α later.) The model is illustrated in Figure 1. *I'm confused about the rationale for relaxing the lower bound on α ?

Model diagram has
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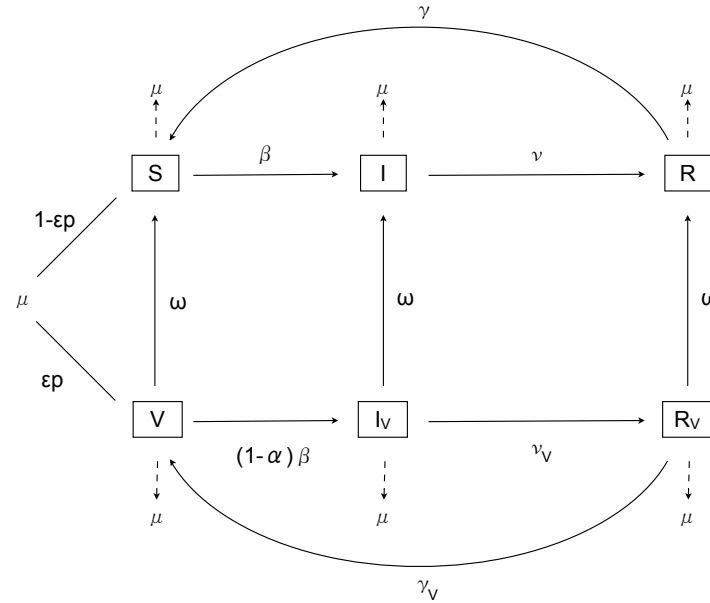


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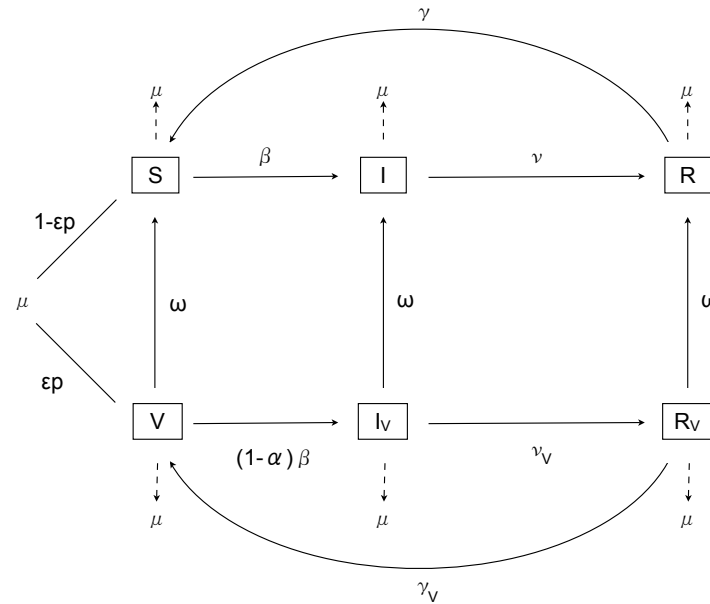


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The first four eigenvalues are always negative. The nontrivial part of characteristic equation satisfies

$$\lambda^2 + b_1\lambda + c_1 = 0,$$

where

$$\begin{aligned} b_1 &= -\beta\bar{S} + \mu + \nu - \beta_V\bar{V} + \nu_V + \mu + \omega \\ c_1 &= (\beta\bar{S} - \mu - \nu)(\beta_V\bar{V} - \nu_V - \mu - \omega) - \beta_V\bar{V}(\beta\bar{S} + \omega) \\ &= \beta\bar{S}(-\nu_V - \mu - \omega) - (\mu + \nu)(\beta_V\bar{V} - \nu_V - \mu - \omega) - \beta_V\bar{V}\omega. \end{aligned}$$

From $c_1 = 0$, we find **I am confused about this step here, sorry! How did we get from setting $c_1 = 0$ to determining R_0 ?**

$$R_0 = \frac{\beta\bar{S}(\nu_V + \mu + \omega) + \beta_V\bar{V}(\mu + \nu + \omega)}{(\mu + \nu)(\mu + \nu_V + \omega)}$$

(This is equivalent to the value found using the next-generation method.)

If $c_1 = 0$ and $b_1 > 0$, then we have a bifurcation with the property that the DFE is stable if $R_0 < 1$ and unstable if $R_0 > 1$

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When $c_1 = 0$, we have

$$b_1 \Big|_{c_1=0} = \frac{1}{\nu_V + \mu + \omega} \left[\beta_V\bar{V}(\nu - \nu_V) + (\nu_V + \mu + \omega)^2 \right].$$

Note that if $\nu = \nu_V$, then $b_1 > 0$. However, it is plausible that vaccinated individuals infected with RSV will recover faster than unvaccinated individuals. Thus $\nu_V > \nu$. This raises the possibility that b_1 could be negative.

If $\nu_V \rightarrow \infty$, then this is equivalent to vaccinated individuals recovering instantaneously. In this case,

$$\begin{aligned} \lim_{\nu_V \rightarrow \infty} b_1 &= \lim_{\nu_V \rightarrow \infty} \frac{\beta_V\bar{V}(\nu - \nu_V)}{\omega + \mu + \nu_V} + \omega + \mu + \nu_V \\ &= \infty - \beta_V\bar{V} > 0 \end{aligned}$$

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Make sure the logic is tight.

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In our model we considered RSV transmission dynamics for a single age class, in order to allow for the model to be analytically tractable. Given we were examining the broad population-level impacts in a large population, we considered this a reasonable model simplification. Further, it has been shown that for a similar compartmental RSV model, including multiple age classes did not change the bifurcation structure of the model [16]. However, different vaccine candidates for RSV are being developed for distinct key age groups – infants, young children, pregnant women, and the elderly [30]. This means that future models that explore the specific implications of vaccines for these target groups may need to incorporate additional age classes.

A vaccine that targets RSV infection has the potential to significantly reduce the overall prevalence of the disease, but it has to be sufficiently long-lasting. Coverage and effectiveness of the vaccine is important, but the critical parameter that our modelling identified is the waning rate of the vaccine. We thus recommend that candidate vaccines be tested for sufficient duration before being released to the public.

References

- [1] Acedo, L., J. Díez-Domingo, J.-A. Morano, and R.-J. Villanueva (2010). Mathematical modelling of respiratory syncytial virus (RSV): vaccination strategies and budget applications. *Epidemiology and Infection* 138(6), 853–60.
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Further implications
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higher than the transmission rate without vaccination. Since we extended the model introduced by Weber *et al.* [39], our model inherited many of the assumptions from that model, such as mass-action transmission, a constant birth rate and that the birth and death rates were matched, resulting in a constant population.

In our model we considered RSV transmission dynamics for a single age class, in order to allow for the model to be analytically tractable. Given we were examining the broad population-level impacts in a large population, we considered this a reasonable model simplification. Further, it has been shown that for a similar compartmental RSV model, including multiple age classes did not change the bifurcation structure of the model [16]. However, different vaccine candidates for RSV are being developed for distinct key age groups – infants, young children, pregnant women, and the elderly [30]. This means that future models that explore the specific implications of vaccines for these target groups may need to incorporate additional age classes.

A vaccine that targets RSV infection has the potential to significantly reduce the overall prevalence of the disease, but it has to be sufficiently long-lasting. Coverage and effectiveness of the vaccine is important, but the critical parameter that our modelling identified is the waning rate of the vaccine. We thus recommend that candidate vaccines be tested for sufficient duration before being released to the public.

References

- [1] Acedo, L., J. Díez-Domingo, J.-A. Morano, and R.-J. Villanueva (2010). Mathematical modelling of respiratory syncytial virus (RSV): vaccination strategies and budget applications. *Epidemiology and Infection* 138(6), 853–60.
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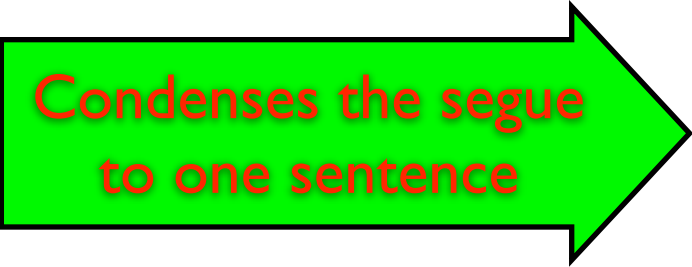
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Here, we examine the effects of a prophylactic vaccine on the transmission of RSV in a single age class. We consider several vaccination scenarios, including differing levels of coverage, seasonal oscillations in the transmission rate and a waning of the vaccine. We also compare continuous vaccination to impulsive vaccination in order to determine conditions on the vaccination strength and duration that will control the virus.

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We extend the [SEIRS](#) compartmental model for a single age cohort described by Weber *et al.* [46] to include a vaccine strategy for RSV where a fixed proportion of newborns are vaccinated [before infection. \(This is equivalent to the situation where pregnant mothers are vaccinated before giving birth.\)](#) We assume that the leaving rate μ is unchanged across all classes and that there is no disease-specific death rate. We scale the entry and leaving rates so that the population is constant.

Let S represent susceptible, I represent infected and R represent recovered individuals, with V , I_V and R_V the corresponding compartments for vaccinated individuals. The birth rate is μ , with a proportion p vaccinated, of whom ϵ successfully mount an immune response; the death rate is equal to the birth rate. The time-dependent transmissibility function is $\beta(t)$, with recovery ν and loss of immunity γ . The transmissibility of infected vaccinated individuals is described by $\beta_V(t)$, and the recovery and loss of immunity rates for vaccinated individuals are ν_V and γ_V respectively. Finally, the waning of the vaccine protectiveness is given by ω .



Condenses the segue
to one sentence

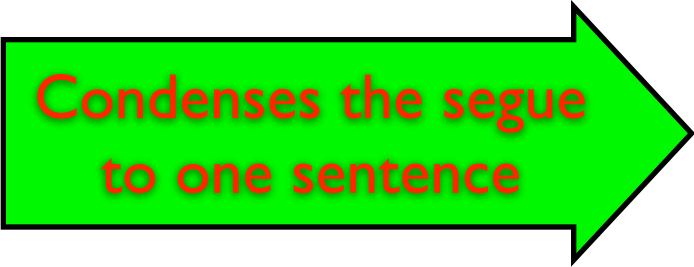
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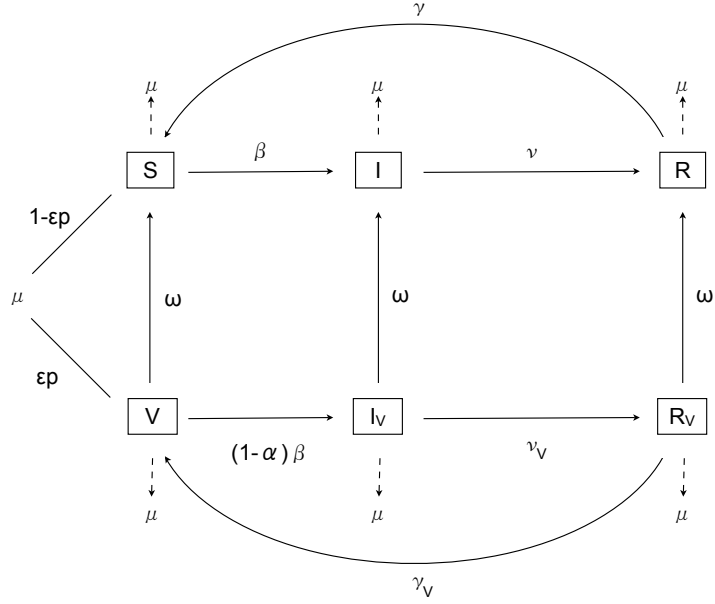


Figure 1: The model.

The basic model with vaccination is then

$$\begin{aligned}
 S' &= \mu(1 - \epsilon p) - \mu S - \beta(t)S(I + I_V) + \gamma R + \omega V \\
 I' &= \beta(t)S(I + I_V) - \nu I - \mu I + \omega I_V \\
 R' &= \nu I - \mu R - \gamma R + \omega R_V \\
 V' &= \epsilon p \mu - \mu V - \beta_V(t)V(I + I_V) + \gamma_V R_V - \omega V \\
 I_V' &= \beta_V(t)V(I + I_V) - \nu_V I_V - \mu I_V - \omega I_V \\
 R_V' &= \nu_V I_V - \mu R_V - \gamma_V R_V - \omega R_V,
 \end{aligned}$$

with $\beta(t) = b_0(1 + b_1 \cos(2\pi t + \phi))$ and $\beta_V(t) = (1 - \alpha)\beta(t)$, for $0 \leq \alpha \leq 1$, where α represents the efficacy of vaccination in preventing infection. (We will relax the lower bound on α later in order to examine some theoretical scenarios.) The model is illustrated in Figure 1.

The first four eigenvalues are always negative. The nontrivial part of characteristic equation satisfies

$$\lambda^2 + b_1\lambda + c_1 = 0,$$

where

$$\begin{aligned} b_1 &= -\beta\bar{S} + \mu + \nu - \beta_V\bar{V} + \nu_V + \mu + \omega \\ c_1 &= (\beta\bar{S} - \mu - \nu)(\beta_V\bar{V} - \nu_V - \mu - \omega) - \beta_V\bar{V}(\beta\bar{S} + \omega) \\ &= \beta\bar{S}(-\nu_V - \mu - \omega) - (\mu + \nu)(\beta_V\bar{V} - \nu_V - \mu - \omega) - \beta_V\bar{V}\omega. \end{aligned}$$

We use the method of the constant term of the characteristic polynomial to determine the reproduction number [19]. Rearranging $c_1 = 0$, we find

$$R_0 = \frac{\beta\bar{S}(\nu_V + \mu + \omega) + \beta_V\bar{V}(\mu + \nu + \omega)}{(\mu + \nu)(\mu + \nu_V + \omega)}$$

(This is equivalent to the value found using the next-generation method.)

If $c_1 = 0$ and $b_1 > 0$, then we have a bifurcation with the property that the DFE is stable if $R_0 < 1$ and unstable if $R_0 > 1$

However, it is possible that when $c_1 = 0$, $b_1 < 0$. In this case, R_0 is not a threshold and the disease can persist if $R_0 < 1$.

When $c_1 = 0$, we have

$$b_1 \Big|_{c_1=0} = \frac{1}{\nu_V + \mu + \omega} \left[\beta_V\bar{V}(\nu - \nu_V) + (\nu_V + \mu + \omega)^2 \right].$$

Note that if $\nu = \nu_V$, then $b_1 > 0$. However, it is plausible that vaccinated individuals infected with RSV will recover faster than unvaccinated individuals. Thus $\nu_V > \nu$. This raises the possibility that b_1 could be negative.

If $\nu_V \rightarrow \infty$, then this is equivalent to vaccinated individuals recovering instantaneously. In this case,

$$\begin{aligned} \lim_{\nu_V \rightarrow \infty} b_1 &= \lim_{\nu_V \rightarrow \infty} \frac{\beta_V\bar{V}(\nu - \nu_V)}{\omega + \mu + \nu_V} + \omega + \mu + \nu_V \\ &= \infty - \beta_V\bar{V} > 0 \end{aligned}$$

Hence if we define $f(\nu_V) = \frac{\beta_V\bar{V}(\nu - \nu_V) + (\omega + \mu + \nu_V)^2}{\omega + \mu + \nu_V}$, then it is clear that $f(0) > 0$ and $f(\infty) > 0$. So we would like to know whether f has a turning point ν_V^* such that $f(\nu_V^*) < 0$.

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Tighten up the chain of logic.

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Unexpected infection spikes in a model of Respiratory Syncytial Virus vaccination

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We thank the Editor and Reviewers for their time and consideration of our manuscript on RSV vaccination. We have done everything the reviewers requested. Here is a point-by-point response to the reviewers.

Reviewer 1

General This reviewer had four major comments and three minor comments.

Response: We have done everything this reviewer suggested. Changes due to this reviewer are in [blue](#).

Comment (1) The authors describe current efforts at vaccine development as “focused on the development of particle-based and subunit vaccines (p.2, 3/4 down)”. They do not mention, or consider, the continuing work on live attenuated vaccines or vectored vaccines both of which are supported by multiple large pharma companies.

Response: Good point. We have changed this sentence to: “focused on the development of particle-based, subunit and vectored vaccines. Live-attenuated vaccines are also undergoing phase 1 trials.” This is based on the summary at <http://www.path.org/vaccineresources/files/RSV-snapshot-December2016.pdf> (Page 2)

Comment (2) The authors consider (p.3, middle) “a vaccine strategy for RSV where a fixed proportion of individuals entering the model are temporarily immune to infection. This reflects the situation where newborn children are vaccinated at birth.” They do not mention, or consider, that the main vaccine strategy now being pursued for the youngest infants is not direct immunization, but is instead maternal immunization. Vaccination occurring during the third trimester generates antibodies in the mother that are transferred transplacentally to the infant, resulting in higher antibody titers in the infant at birth. The thought is that the higher antibody titers should protect the infant for approximately two months longer. Pre-formed antibodies decay with time, and by 6 months maternal antibodies are no longer detectable

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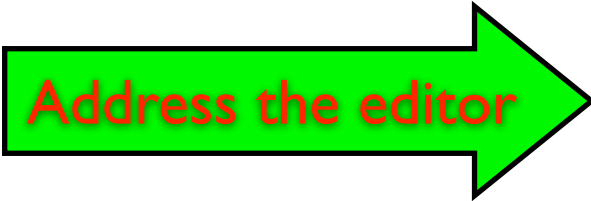
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Indicate what you've changed
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Response: This is an excellent point, so we have changed the focus in this section to maternal vaccination and discussed this in some detail. Happily, by doing so, the results are unchanged from a mathematical perspective. (Pages 3, 7, 16, 17)

Comment (3) Vaccination of infants as soon as they are born is seldom successful for any pathogen because the infants immune system is immature. It is not currently being contemplated for RSV. However, MedImmune (owner of the prophylactic monoclonal antibody that is currently used for “at risk” infants to protect them against RSV) has developed a more potent RSV-neutralizing monoclonal antibody with increased stability that could be given at birth to protect infants for their first 6 months. This approach would avoid the uncertainties of individual maternal responses to RSV and the problem of premature birth which could result in incomplete transfer of the antibodies elicited by a maternal vaccine, depending on the time of vaccination relative to birth.

In general, the thinking in the field is that there will be two vaccines for RSV, one to protect infants during their first 6 months, and another to protect them from 6 months on. I realize that there may be too many variables for the authors to consider in one report, but they could choose one of these strategies and model that. Maternal vaccination before birth would seem to be the most important to study now since it is being pursued aggressively by the NIH and two big pharma companies and several smaller companies, and is being supported by the Bill and Melinda Gates Foundation. The MedImmune stabilized monoclonal antibody approach could be included as generally equivalent.

Response: This brings up a point that we realise was not clear: we are actually considering both options. The nonimpulsive model considers pre-infection vaccination only, while the impulsive model considers subsequent vaccination. We have added emphasis in several places to make this clear. (Pages 3, 7, 11, 16, 17)

Comment (4) But the protection of any of these approaches would cover only the first 6 months of life. Thereafter, immunization of the child with another vaccine would be needed to induce active immunity and a recall response that would provide future, more rapid protection upon infection. Right now, live attenuated or vectored (adenovirus) vaccines are the front runners, but direct immunization with a subunit vaccine might eventually be considered. A subunit vaccine has not been considered largely because of the initial formalin-inactivated vaccine trial in the 1960s resulted in much more severe disease following the first community acquired infection in the vaccinees.

While modeling a 10-year protective vaccine and a lifetime-protective vaccine can be done, even infection with the wild-type virus does not provide 10-year protection, so it is difficult to see how a long-term protective vaccine could be generated. Nevertheless, it is a laudable goal.

Response: This is a helpful observation. We have decided to change our focus away from long-lasting vaccines and instead mostly focus on short-term durations, as the

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In general, the thinking in the field is that there will be two vaccines for RSV, one to protect infants during their first 6 months, and another to protect them from 6 months on. I realize that there may be too many variables for the authors to consider in one report, but they could choose one of these strategies and model that. Maternal vaccination before birth would seem to be the most important to study now since it is being pursued aggressively by the NIH and two big pharma companies and several smaller companies, and is being supported by the Bill and Melinda Gates Foundation. The MedImmune stabilized monoclonal antibody approach could be included as generally equivalent.

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While modeling a 10-year protective vaccine and a lifetime-protective vaccine can be done, even infection with the wild-type virus does not provide 10-year protection, so it is difficult to see how a long-term protective vaccine could be generated. Nevertheless, it is a laudable goal.

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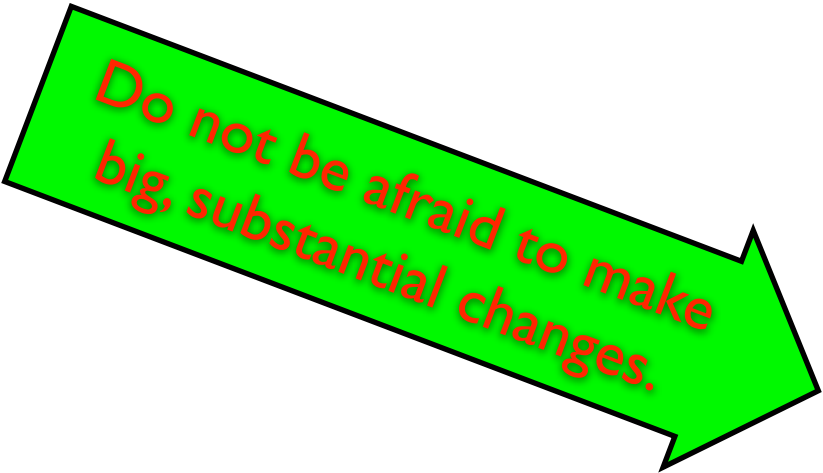
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Comment (5) p.1, author list. Why is Robert J. Smith followed by a ? ?

Response: It is part of the author's name. See, for example:
<http://mysite.science.uottawa.ca/rsmith43/MDRHIV.pdf>

Comment (6) p.18, l.10. vaccination-induced

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Response: We agree, although this sentence has now been deleted, so it no longer applies.

Reviewer 2

General This reviewer noted that the research questions examined in our manuscript are extremely important and relevant and that we use an innovative approach to address the question of potential vaccine efficacy. This reviewer had five major comments.

Response: We have done everything this reviewer suggested. Changes due to this reviewer are in red.

Comment (1) The authors base the model on the assumption that infants will be given the vaccine at birth. While this is true for a few vaccines, most are not given at birth. Additionally, the most advanced vaccines in development are not being targeted to infants. They are primarily targeting the elderly, and pregnant mothers to protect newborns. The authors need to address the fact that their assumption is very unlikely, or even false more than they have as the manuscript stands.

Response: This is a good point that was also raised by Reviewer #1. See our response to Comments (2) and (3) above. Note in particular that we are actually considering both and have made that more clear. (Pages 3, 12, 16, 17, in blue.)

Comment (2) The authors conclude that vaccine duration would be more important than vaccine coverage. They recommend that vaccine be tested for duration before approval. The authors need to discuss how the practicality of studying long term immunity is very challenging, especially regarding the time frames they test. Obviously 70, or even 10 years would be impossible to test during a clinical trial before licensure.

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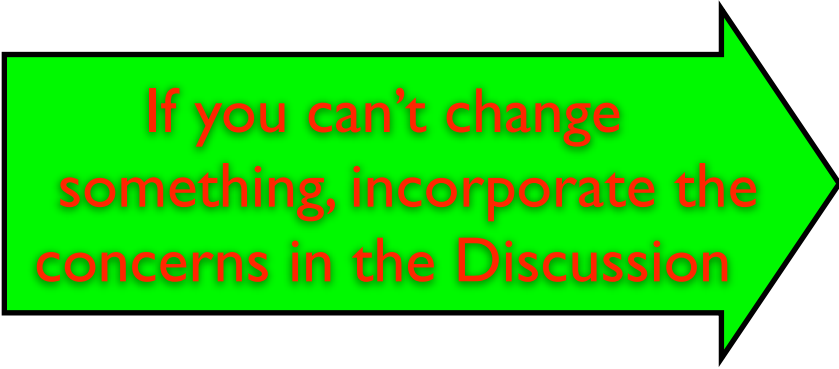
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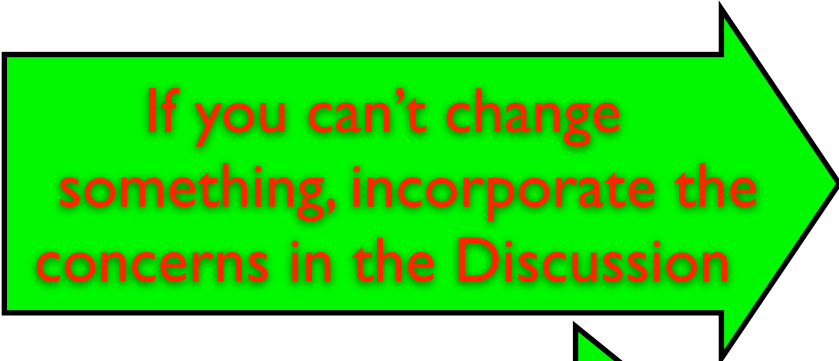
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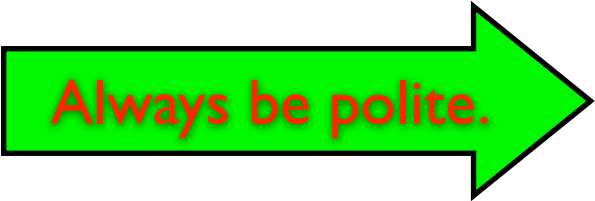
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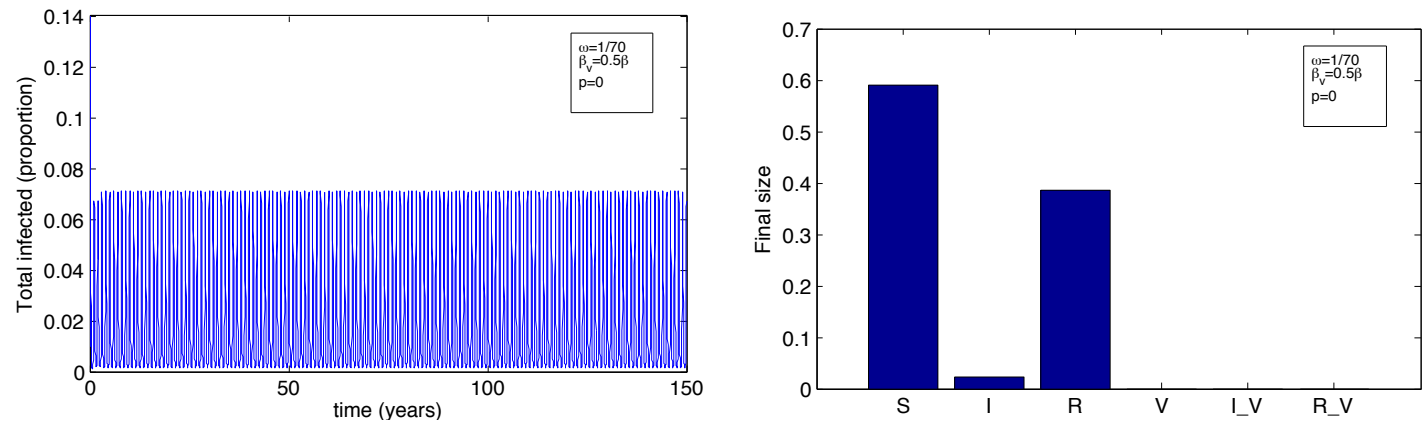


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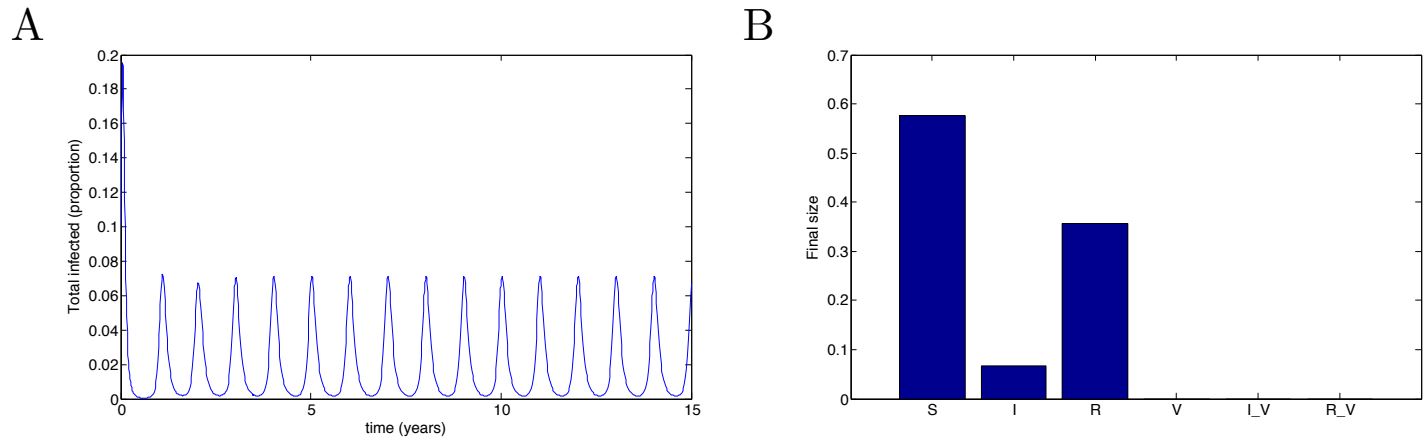


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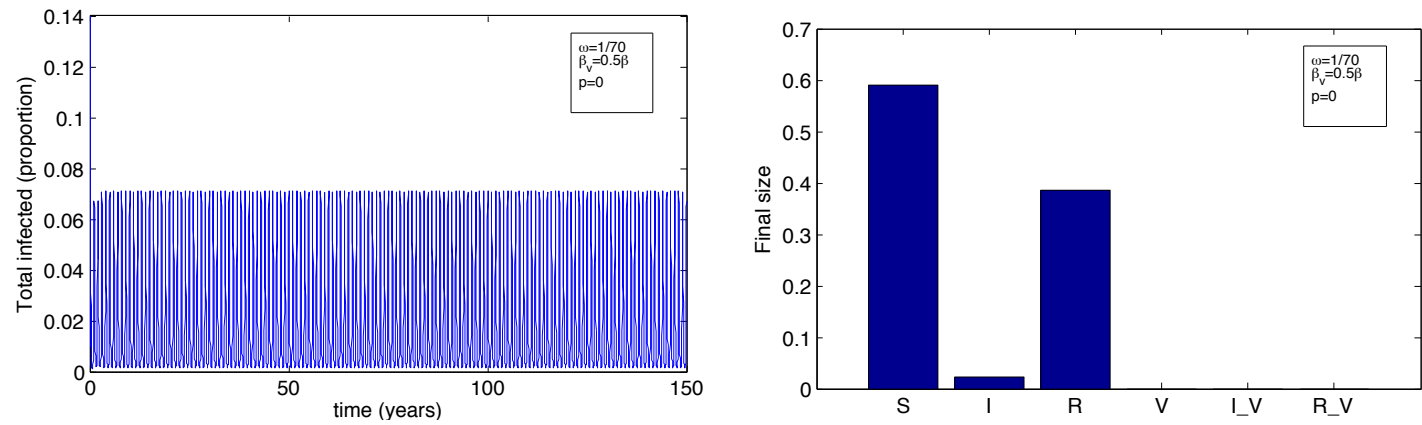


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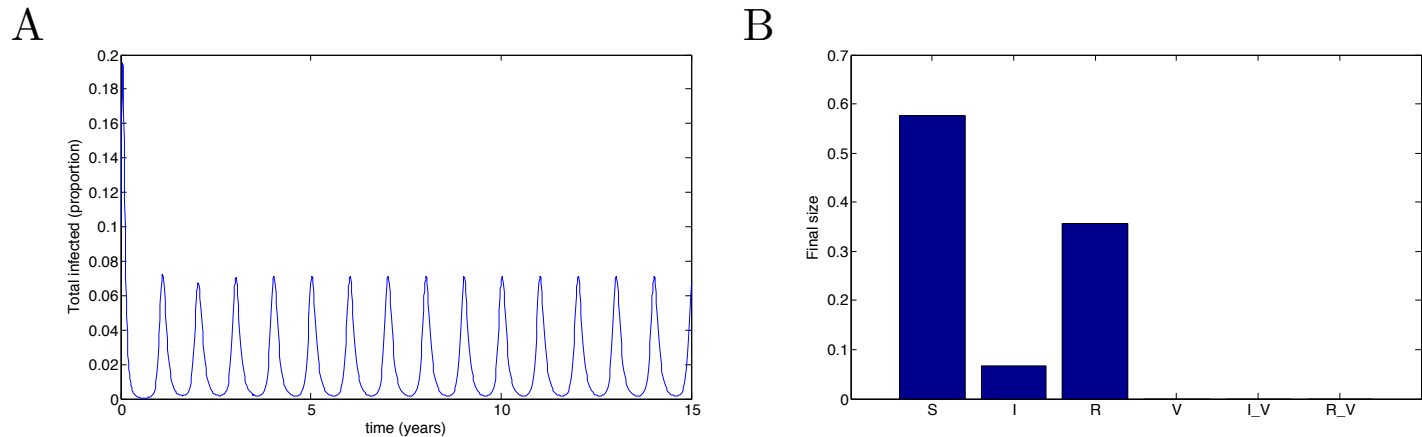


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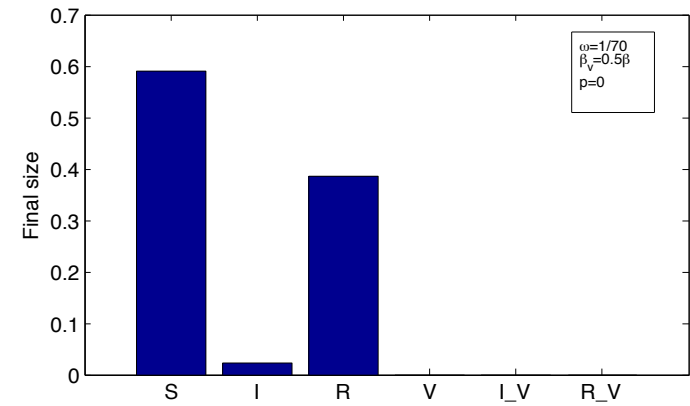
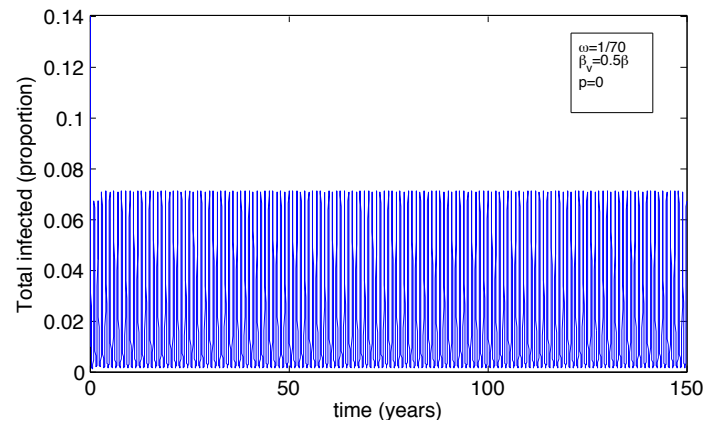
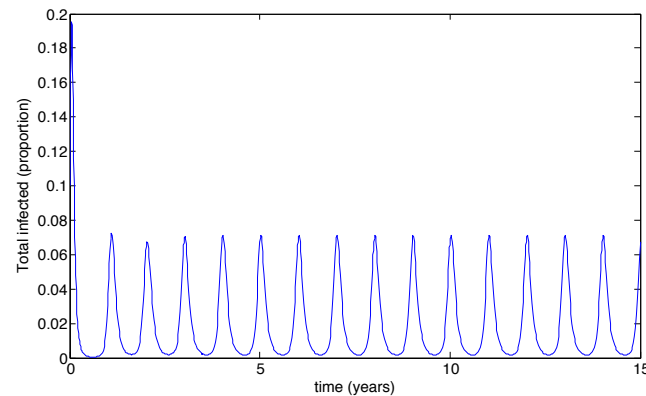


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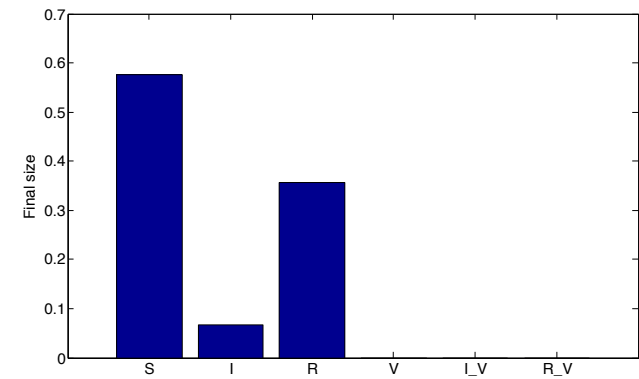


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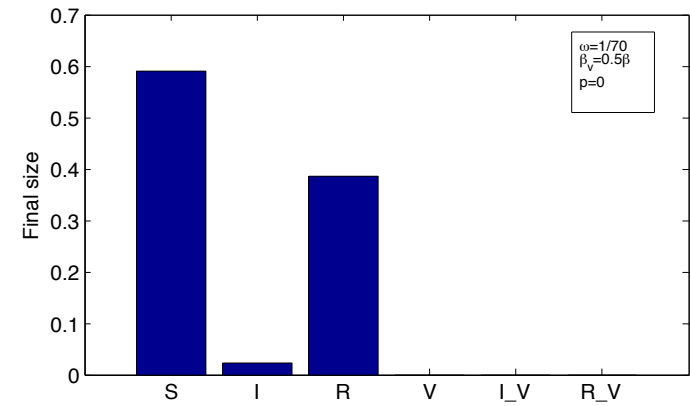
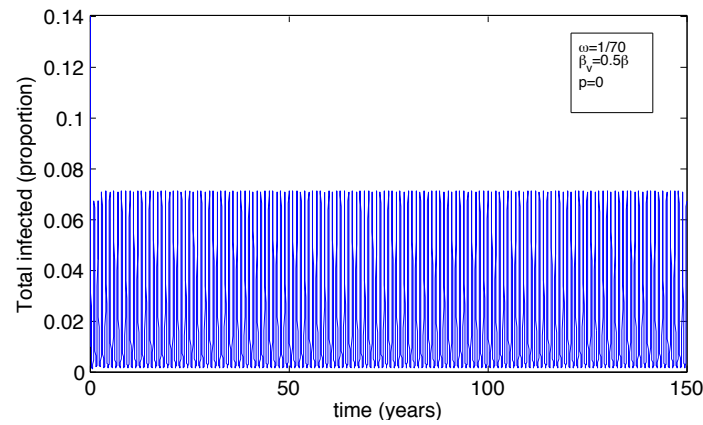
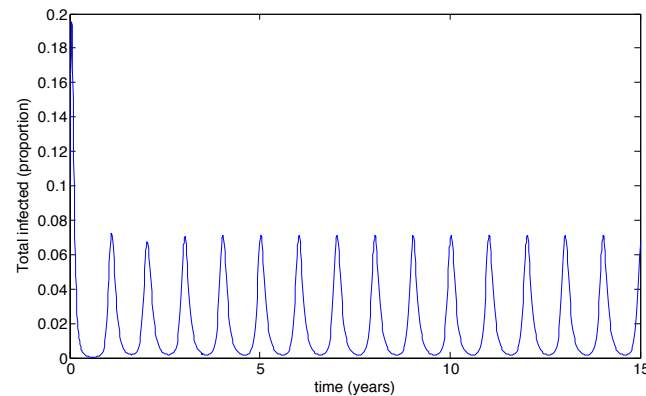


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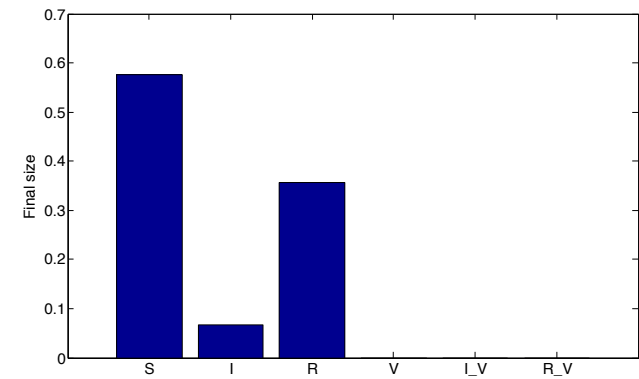


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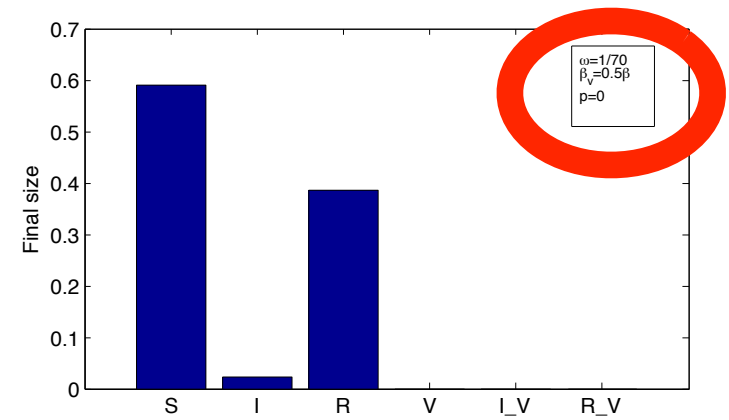
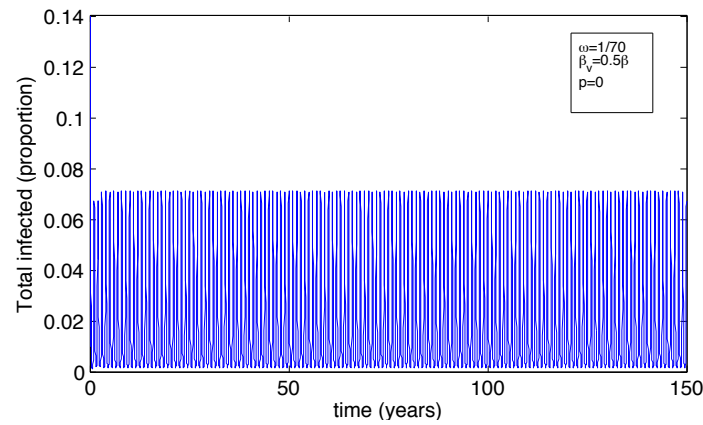
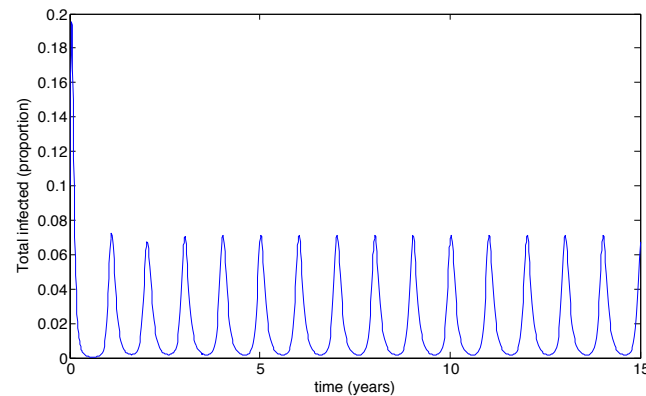


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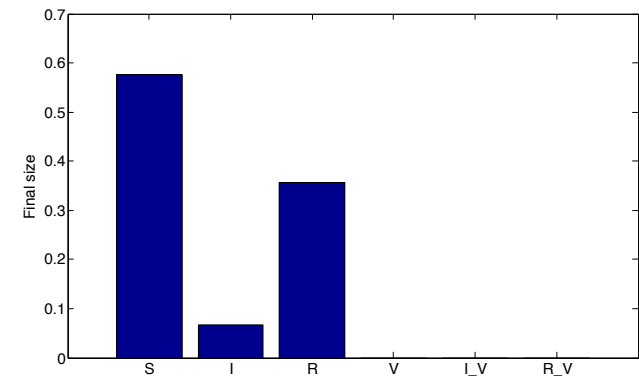


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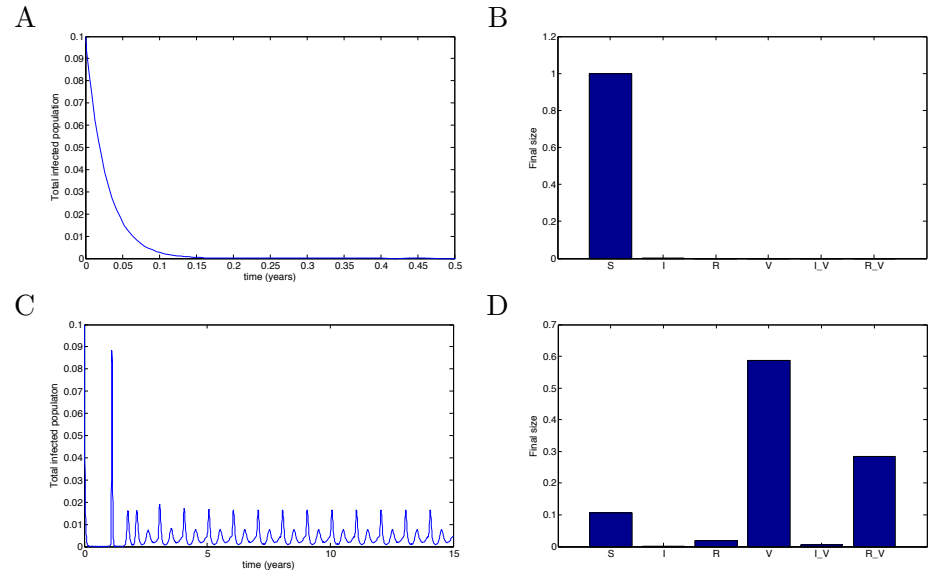
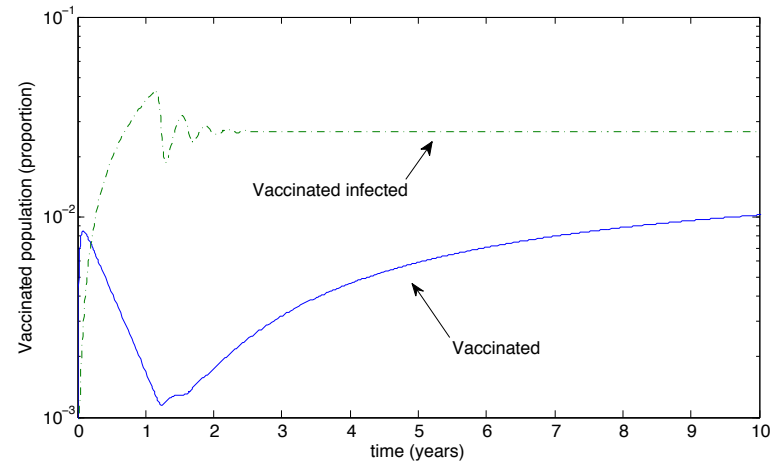


Figure 8: Extreme parameters show that perfect vaccination can induce unexpected infection spikes. A. With no vaccine ($r = 0$), the result is that the infection clears and the entire population remains susceptible. (Note that the timescale is given for only 0.5 years to show the decline but was run for 15 years.) B. The final size of each compartment in the case of no vaccine after 15 years. C. When an imperfect vaccine is given to the entire population ($r = 1$), the result is a series of **disease spikes in the vaccinated population**. Note that the transmission rate is not oscillating in this example. D. The final size of each compartment in the case of full vaccination after 15 years. Vaccination thus destabilises the DFE.

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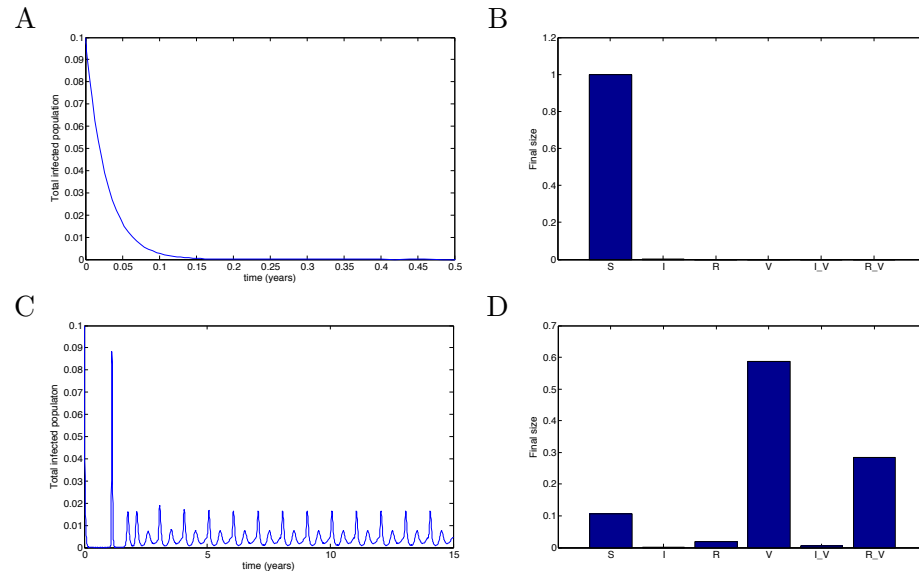
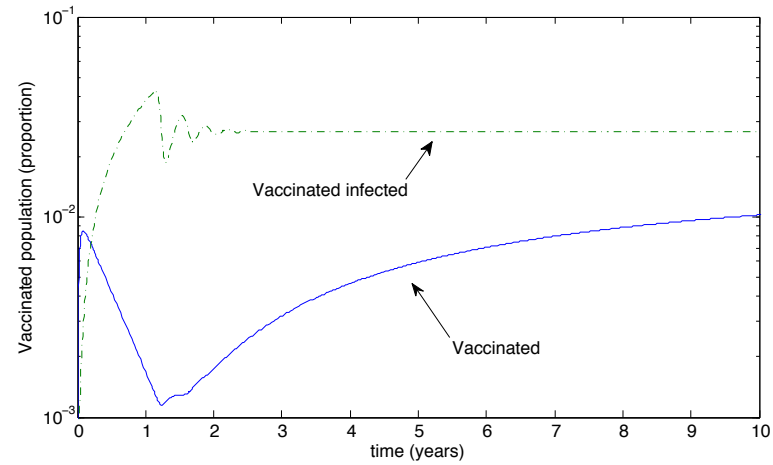
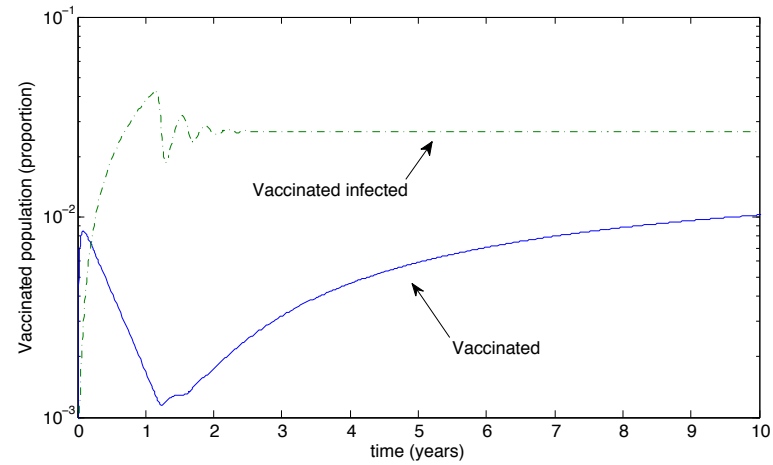


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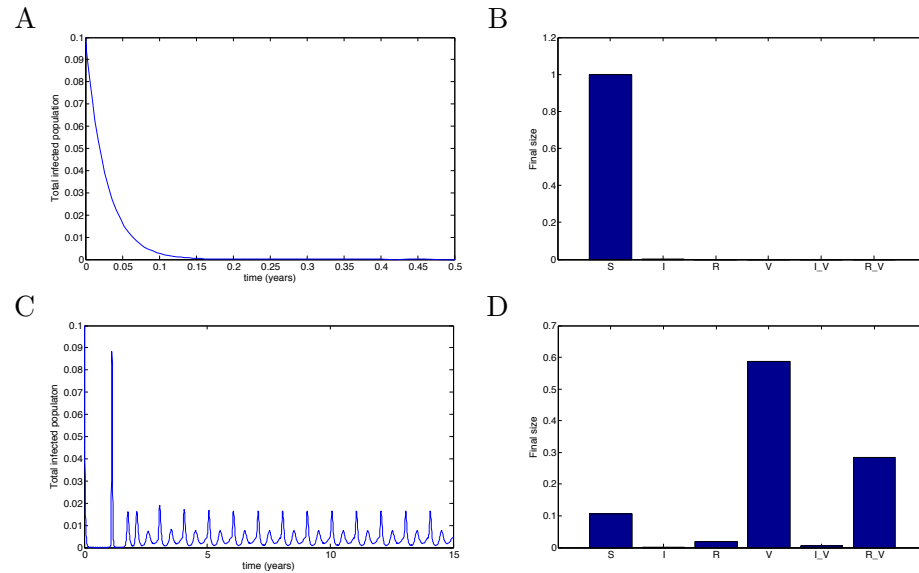


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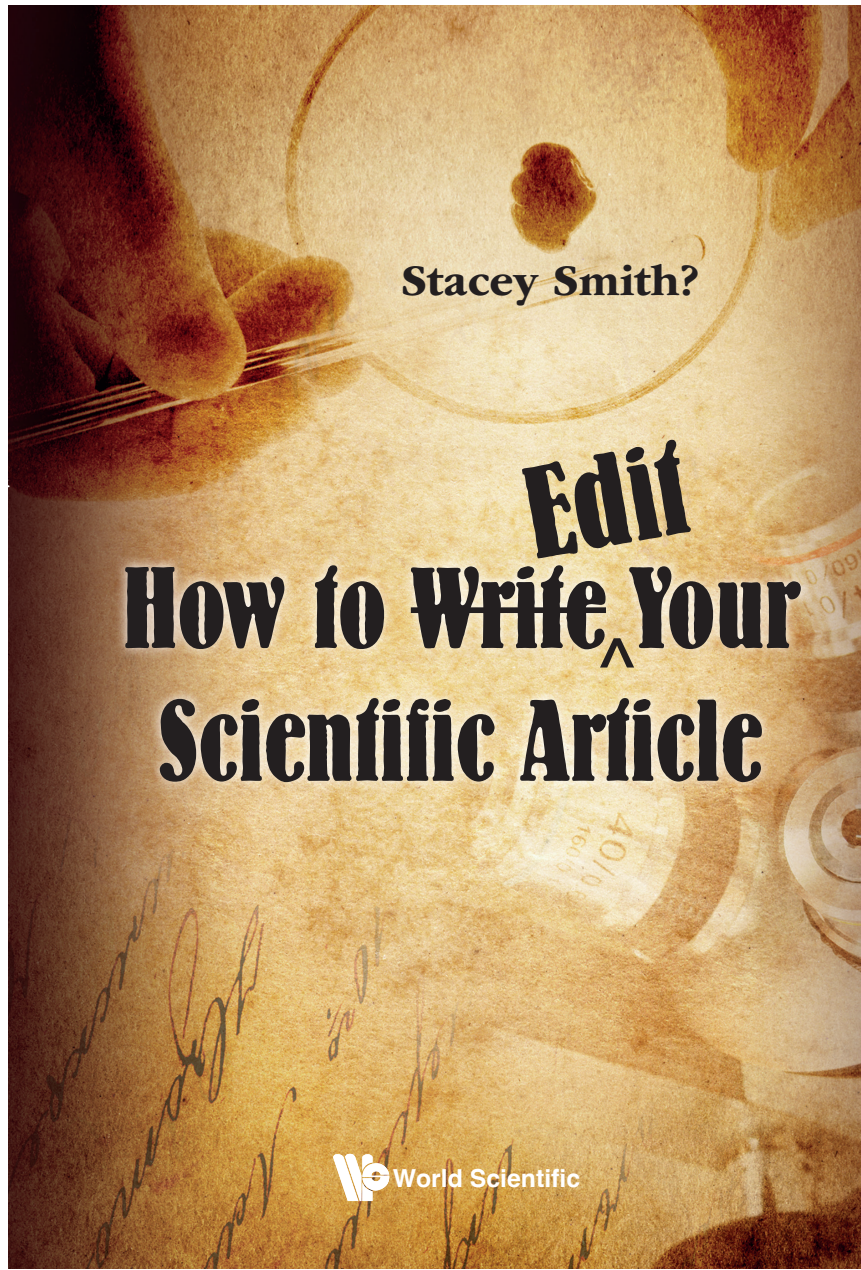
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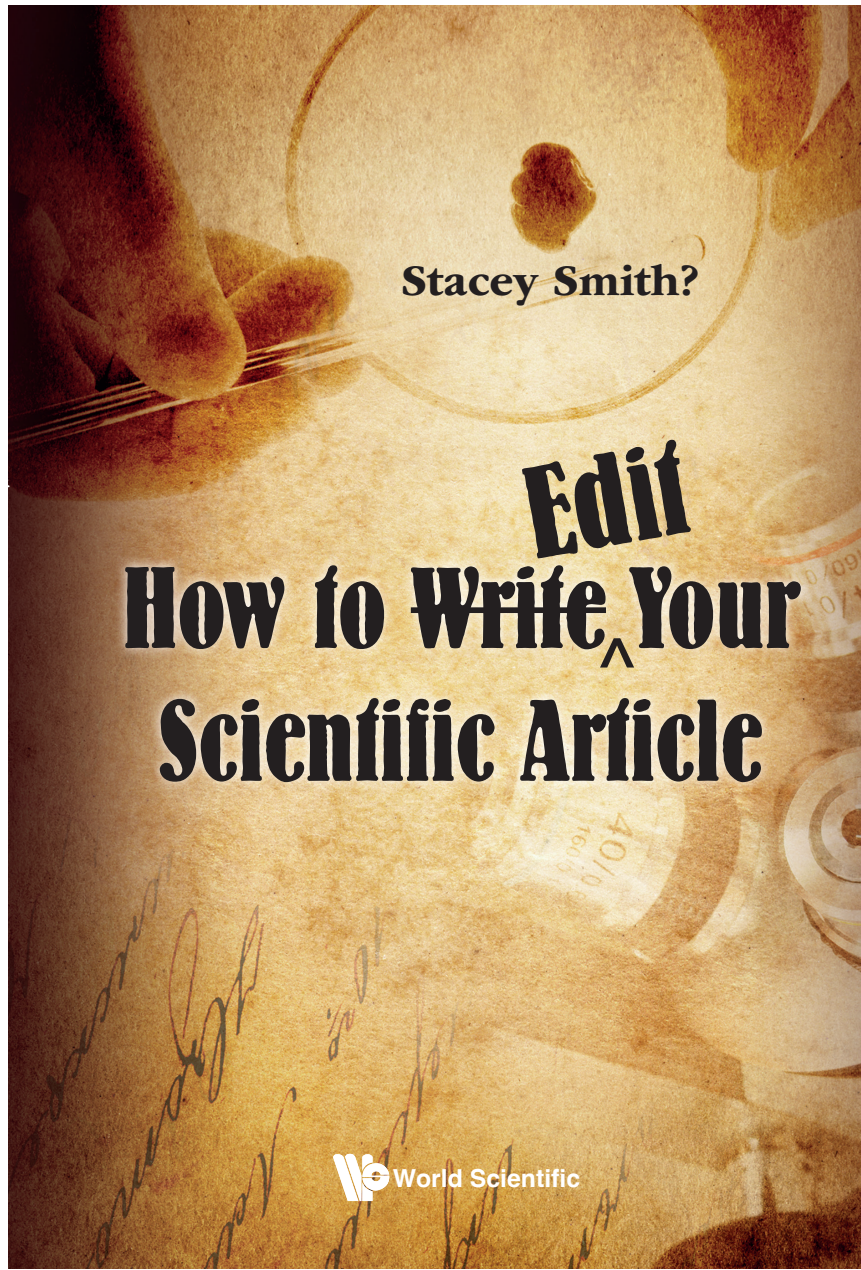
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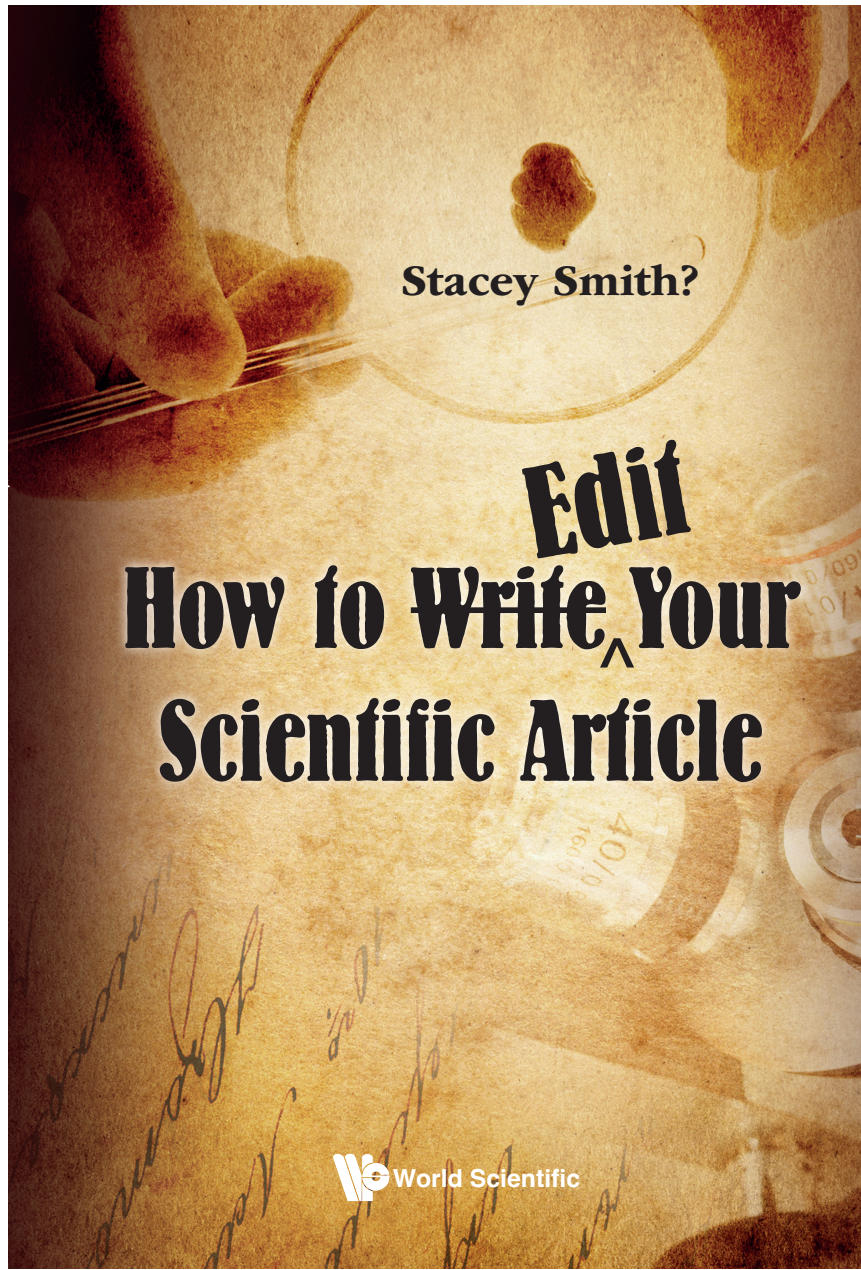


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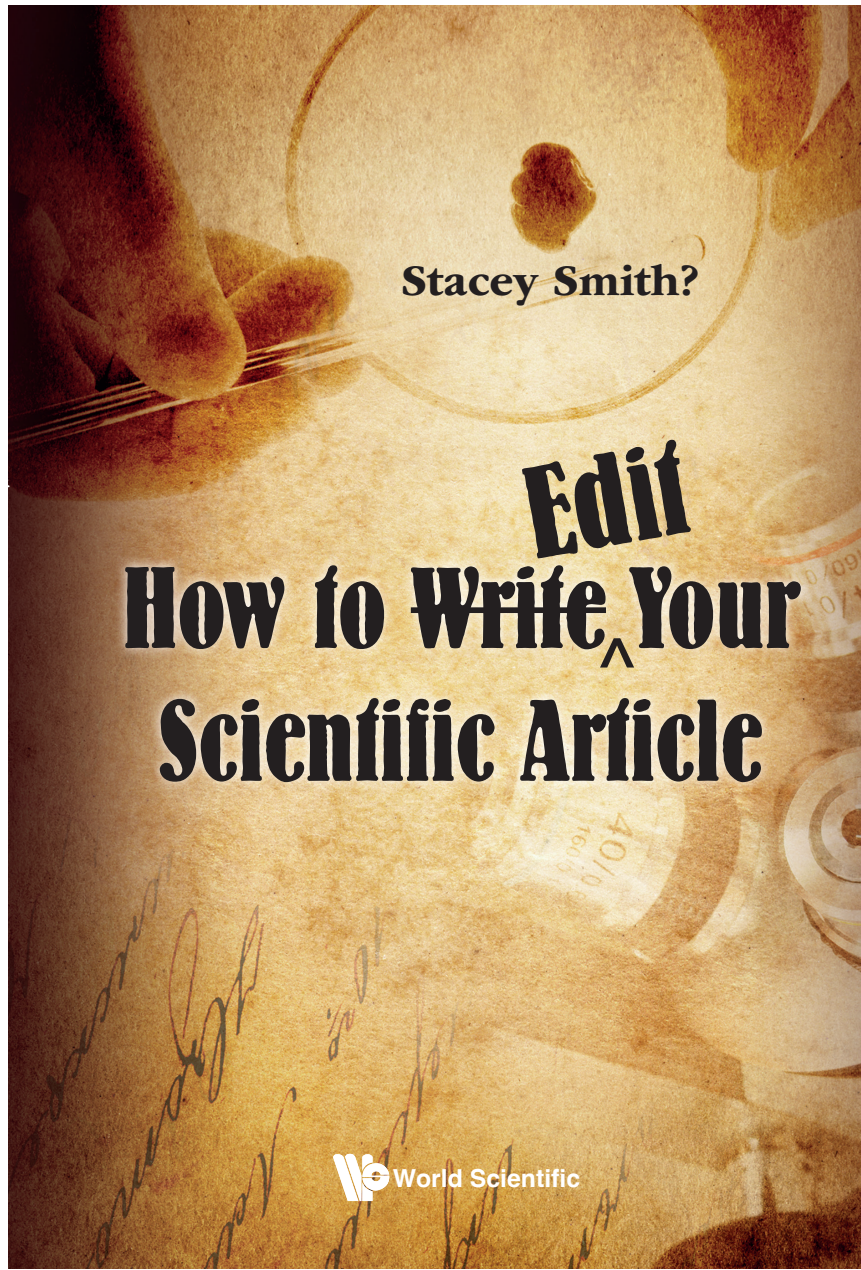
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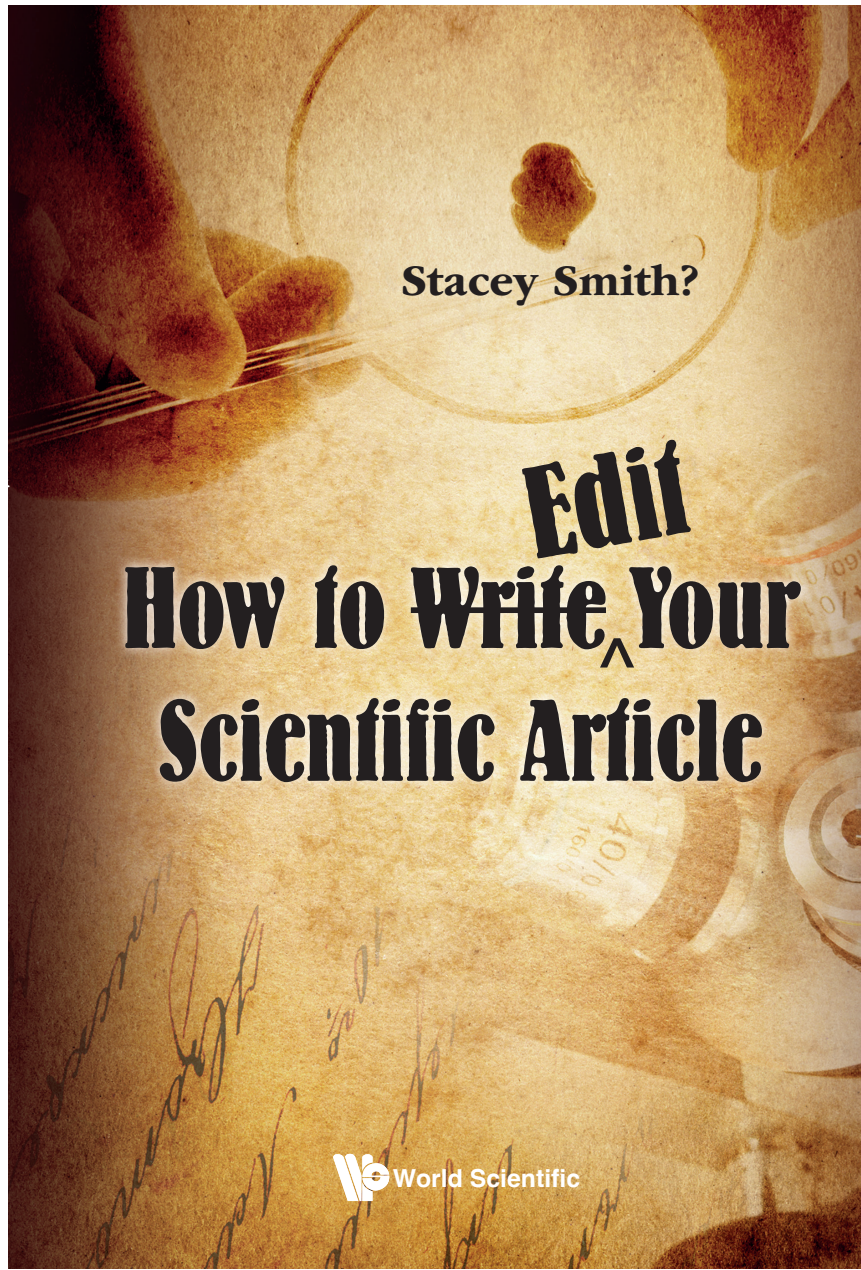
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Includes:

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- what not to do



Further reading



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Includes:

- worked examples
- what not to do
- writing exercises.



The Handout

rsmith43.github.io/mysite/paperwritinghandouts.htm



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Includes links to:



The Handout

rsmith43.github.io/mysite/paperwritinghandouts.htm

Includes links to:

- slides from this talk



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Includes links to:

- slides from this talk
- multiple drafts
- biological introduction
- response to reviewers
- final version.

